

# Logs

## Capítulo 4

sbu: 11 minutos 9,355 segundos

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## Capítulo 5

-Capitulo 5.2: Binutils-2.45 – Pass 1 hecho a las 15:30

Comandos principales ejecutados: mkdir, cd, configure, make, make install.

Obs: tuve que volver a preparar Binutils pq no le puse espacio antes de cada \. No tengo idea de como verificar si esta todo en orden.

```
make[4]: Leaving directory '/mnt/lfs/sources/binutils-2.45/build/ld'
make[3]: Leaving directory '/mnt/lfs/sources/binutils-2.45/build/ld'
make[2]: Leaving directory '/mnt/lfs/sources/binutils-2.45/build/ld'
make[1]: Nothing to be done for 'all-target'.
make[1]: Leaving directory '/mnt/lfs/sources/binutils-2.45/build'
lfs:/mnt/lfs/sources/binutils-2.45/build$
```

-Capitulo 5.3: GCC-15.2.0 – Pass 1 hecho a las 18:45

Comandos principales ejecutados: tar, mv, sed, mkdir, cd, configure, make, make install.

```
make[2]: Leaving directory '/mnt/lfs/sources/gcc-15.2.0/build/x86_64-lfs-linux-gnu/libgcc'
make[1]: Leaving directory '/mnt/lfs/sources/gcc-15.2.0/build'
lfs:/mnt/lfs/sources/gcc-15.2.0/build$ cd ..
lfs:/mnt/lfs/sources/gcc-15.2.0$ cat gcc/limitx.h gcc/glimits.h gcc/limity.h > \
> `dirname ${LFS_TGT-gcc -print-libgcc-file-name}`/include/limits.h
lfs:/mnt/lfs/sources/gcc-15.2.0$
make[2]: Leaving directory '/mnt/lfs/sources/gcc-15.2.0/build/x86_64-lfs-linux-gnu/libgcc'
make[1]: Leaving directory '/mnt/lfs/sources/gcc-15.2.0/build'
lfs:/mnt/lfs/sources/gcc-15.2.0/build$
```

-Capitulo 5.4: Linux-6.16.1 API Headers hecho a las 18:55

Comandos principales ejecutados: make mrproper, make, find, cd.

-Capitulo 5.5: Glibc-2.42 hecho a las 20:00

Comandos principales ejecutados: uname, ln, patch, mkdir, cd, echo, configure, make, make DESTDIR=\$LFS install, sed, grep, rm.

Obs: tardó una hora.

```
lfs:/mnt/lfs/sources/glibc-2.42/build$ echo 'int main(){}' | $LFS_TGT-gcc -x c - -v -Wl,--verbose &> dummy.log
lfs:/mnt/lfs/sources/glibc-2.42/build$ grep -E -o "$LFS/lib.*?cert[1in].*succeeded" dummy.log
/mnt/lfs/lib/./lib/Sert1.o succeeded
/mnt/lfs/lib/./lib/crt1.o succeeded
/mnt/lfs/lib/./lib/crtn.o succeeded
lfs:/mnt/lfs/sources/glibc-2.42/build$ grep -B3 '^ $LFS/usr/include" dummy.log
#include <...> search starts here:
/mnt/lfs/tools/lib/gcc/x86_64-lfs-linux-gnu/15.2.0/include
/mnt/lfs/tools/lib/gcc/x86_64-lfs-linux-gnu/15.2.0/include-fixed
/mnt/lfs/usr/include
lfs:/mnt/lfs/sources/glibc-2.42/build$ grep 'SEARCH.*usr/lib' dummy.log | sed 's|;|\\nlg'
SEARCH_DIR("/mnt/lfs/tools/x86_64-lfs-linux-gnu/lib64")
SEARCH_DIR("/usr/local/lib64")
SEARCH_DIR("/lib64")
SEARCH_DIR("/usr/lib64")
SEARCH_DIR("/mnt/lfs/tools/x86_64-lfs-linux-gnu/lib")
SEARCH_DIR("/usr/local/lib")
SEARCH_DIR("/lib")
SEARCH_DIR("/usr/lib");
lfs:/mnt/lfs/sources/glibc-2.42/build$ grep "/lib.*libc.so.6 " dummy.log
attempt to open /mnt/lfs/usr/lib/libc.so.6 succeeded
lfs:/mnt/lfs/sources/glibc-2.42/build$ grep found dummy.log
found ld-linux-x86-64.so.2 at /mnt/lfs/usr/lib/ld-linux-x86-64.so.2
lfs:/mnt/lfs/sources/glibc-2.42/build$ rm -v a.out dummy.log
removed 'a.out'
removed 'dummy.log'
lfs:/mnt/lfs/sources/glibc-2.42/build$
```

-Capítulo 5.6: Libstdc++ from GCC-15.2.0 hecho a las 20:30

Comandos principales ejecutados: mkdir, cd, configure, make, make DESTDIR=\$LFS install, rm.

```
lfs:/mnt/lfs/sources/gcc-15.2.0/build-libstdc++$ rm -v $LFS/usr/lib/lib{stdc++{,exp,fs},supc++}.la
removed '/mnt/lfs/usr/lib/libstdc++.la'
removed '/mnt/lfs/usr/lib/libstdc++exp.la'
removed '/mnt/lfs/usr/lib/libstdc++fs.la'
removed '/mnt/lfs/usr/lib/libsupc++.la'
lfs:/mnt/lfs/sources/gcc-15.2.0/build-libstdc++$
```

## Capítulo 6

-Capítulo 6.2 hecho a las 23:02

- Instalacion de paquete M4, que contiene un procesador de macros.
- Comandos principales ejecutados: configure, make, make DESTDIR=\$LFS install.

```
make[2]: Leaving directory '/mnt/lfs/sources/m4-1.4.20/po'
Making install in tests
make[2]: Entering directory '/mnt/lfs/sources/m4-1.4.20/tests'
make install-recursive
make[3]: Entering directory '/mnt/lfs/sources/m4-1.4.20/tests'
Making install in .
make[4]: Entering directory '/mnt/lfs/sources/m4-1.4.20/tests'
make[5]: Entering directory '/mnt/lfs/sources/m4-1.4.20/tests'
make[5]: Nothing to be done for 'install-exec-am'.
make[5]: Leaving directory '/mnt/lfs/sources/m4-1.4.20/tests'
make[4]: Leaving directory '/mnt/lfs/sources/m4-1.4.20/tests'
make[3]: Leaving directory '/mnt/lfs/sources/m4-1.4.20/tests'
make[2]: Leaving directory '/mnt/lfs/sources/m4-1.4.20/tests'
lfs:/mnt/lfs/sources/m4-1.4.20$
```

-Capítulo 6.3 hecho a las 23:35

- Instalación de paquete Ncurses-6.5-20250809, que contiene bibliotecas para el manejo de pantallas de caracteres de forma independiente del terminal.
- Comandos principales ejecutados: mkdir, pushd, configure, make, install, popd, make DESTDIR=\$LFS install, ln, sed.

```
make11: Leaving directory '/mnt/lfs/sources/ncurses-6.5-20250809/c++'
lfs:/mnt/lfs/sources/ncurses-6.5-20250809$ ln -sv libncursesw.so $LFS/usr/lib/libncurses.so
'/mnt/lfs/usr/lib/libncurses.so' -> 'libncursesw.so'
lfs:/mnt/lfs/sources/ncurses-6.5-20250809$ sed -e 's/^#if.*XOPEN.*$/#if 1/' \
> -i $LFS/usr/include/curses.h
```

-Capítulo 6.4 hecho a las 23:48

- Instalación del paquete Bash-5.3, que contiene la Bourne-Again Shell.
- Comandos principales ejecutados: configure, make, make DESTDIR=\$LFS install, ln.

```
make11: Leaving directory '/mnt/lfs/sources/bash-5.3/examples/loadables'
lfs:/mnt/lfs/sources/bash-5.3$ ln -sv bash $LFS/bin/sh
'/mnt/lfs/bin/sh' -> 'bash'
lfs:/mnt/lfs/sources/bash-5.3$
```

-Capitulo 6.5 hecho a las 23:59

- Instalación del paquete Coreutils-9.7, que contiene los programas utilitarios básicos necesarios para todo sistema operativo.
- Comandos principales ejecutados: configure, make, make DESTDIR=\$LFS install, mv, mkdir, sed.

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-Capítulo 6.6 hecho a las 9:10

- Instalación del paquete Diffutils-3.12, que contiene programas que muestran las diferencias entre archivos o directorios.
- Comandos principales ejecutados: configure, make, make DESTDIR=\$LFS install.

-Capitulo 6.7 hecho a las 9:20

- Instalación del paquete File-5.46, que contiene una utilidad para determinar el tipo de uno o varios archivos dados.
- Comandos principales ejecutados: mkdir, pushd, popd, configure, make, DESTDIR=\$LFS install, rm.

-Capitulo 6.8 hecho a las 9:30

- Instalación del paquete Findutils-4.10.0, que contiene programas para encontrar archivos. Se incluyen programas para buscar a través de todos los archivos en un árbol de directorios y para crear, mantener y consultar una base de datos (a menudo más rápido que la búsqueda recursiva con find, pero poco fiable si la base de datos no se ha actualizado recientemente). Findutils también proporciona el programa xargs, que puede usarse para ejecutar un comando específico sobre cada archivo seleccionado por una búsqueda.
- Comandos principales ejecutados: configure, make, make DESTDIR=\$LFS install.

-Capitulo 6.9 hecho a las 9:40.

- Instalación del paquete Gawk-5.3.2, que contiene programas para manipular archivos de texto.
- Comandos principales ejecutados: sed, configure, make, make DESTDIR=\$LFS install.

```

make[2]: Leaving directory '/mnt/lfs/sources/gawk-5.3.2/po'
Making install in test
make[2]: Entering directory '/mnt/lfs/sources/gawk-5.3.2/test'
make[3]: Entering directory '/mnt/lfs/sources/gawk-5.3.2/test'
make[3]: Nothing to be done for 'install-exec-am'.
make[3]: Nothing to be done for 'install-data-am'.
make[3]: Leaving directory '/mnt/lfs/sources/gawk-5.3.2/test'
make[2]: Leaving directory '/mnt/lfs/sources/gawk-5.3.2/test'
make[1]: Leaving directory '/mnt/lfs/sources/gawk-5.3.2'
lfs:/mnt/lfs/sources/gawk-5.3.2$ cd ..
lfs:/mnt/lfs/sources$

```

-Capítulo 6.10 hecho a las 9:50.

- Instalación del paquete Grep-3.12, que contiene programas para buscar dentro del contenido de archivos.
- Comandos principales ejecutados: configure, make, make DESTDIR=\$LFS install.

```

make[4]: Leaving directory '/mnt/lfs/sources/grep-3.12/gnulib-tests'
make[3]: Leaving directory '/mnt/lfs/sources/grep-3.12/gnulib-tests'
make[2]: Leaving directory '/mnt/lfs/sources/grep-3.12/gnulib-tests'
make[1]: Leaving directory '/mnt/lfs/sources/grep-3.12/gnulib-tests'
make[1]: Entering directory '/mnt/lfs/sources/grep-3.12'
make[2]: Entering directory '/mnt/lfs/sources/grep-3.12'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/mnt/lfs/sources/grep-3.12'
make[1]: Leaving directory '/mnt/lfs/sources/grep-3.12'
lfs:/mnt/lfs/sources/grep-3.12$ cd ..

```

-Capítulo 6.11 hecho a las 10:00.

- Instalación del paquete Gzip-1.14, que contiene programas para comprimir y descomprimir archivos.
- Comandos principales ejecutados: configure, make, make DESTDIR=\$LFS install.

```

make[3]: Leaving directory '/mnt/lfs/sources/gzip-1.14'
make[2]: Leaving directory '/mnt/lfs/sources/gzip-1.14'
Making install in tests
make[2]: Entering directory '/mnt/lfs/sources/gzip-1.14/tests'
make[3]: Entering directory '/mnt/lfs/sources/gzip-1.14/tests'
make[3]: Nothing to be done for 'install-exec-am'.
make[3]: Nothing to be done for 'install-data-am'.
make[3]: Leaving directory '/mnt/lfs/sources/gzip-1.14/tests'
make[2]: Leaving directory '/mnt/lfs/sources/gzip-1.14/tests'
make[1]: Leaving directory '/mnt/lfs/sources/gzip-1.14'
lfs:/mnt/lfs/sources/gzip-1.14$

```

-Capítulo 6.12 hecho a las 10:07.

- Instalación del paquete Make-4.4.1, que contiene un programa para controlar la generación de ejecutables y otros archivos que no son de código fuente de un paquete a partir de archivos fuente.
- Comandos principales ejecutados: configure, make, make DESTDIR=\$LFS install.

```

make[2]: Leaving directory '/mnt/lfs/sources/make-4.4.1/doc'
make[1]: Leaving directory '/mnt/lfs/sources/make-4.4.1/doc'
make[1]: Entering directory '/mnt/lfs/sources/make-4.4.1'
make[2]: Entering directory '/mnt/lfs/sources/make-4.4.1'
/usr/bin/mkdir -p '/mnt/lfs/usr/bin'
/usr/bin/install -c make '/mnt/lfs/usr/bin'
/usr/bin/mkdir -p '/mnt/lfs/usr/include'
/usr/bin/install -c -m 644 src/gnumake.h '/mnt/lfs/usr/include'
/usr/bin/mkdir -p '/mnt/lfs/usr/share/man/man1'
/usr/bin/install -c -m 644 doc/make.1 '/mnt/lfs/usr/share/man/man1'
make[2]: Leaving directory '/mnt/lfs/sources/make-4.4.1'
make[1]: Leaving directory '/mnt/lfs/sources/make-4.4.1'
lfs:/mnt/lfs/sources/make-4.4.1$

```

-Capítulo 6.13 hecho a las 10:15.

- Instalación del paquete Patch-2.8, que contiene un programa para modificar o crear archivos aplicando un archivo «patch», normalmente generado por el programa diff.
- Comandos principales ejecutados: configure, make, make DESTDIR=\$LFS install.

```

make[3]: Leaving directory '/mnt/lfs/sources/patch-2.8/tests'
make[2]: Leaving directory '/mnt/lfs/sources/patch-2.8/tests'
make[2]: Entering directory '/mnt/lfs/sources/patch-2.8'
make[3]: Entering directory '/mnt/lfs/sources/patch-2.8'
make[3]: Nothing to be done for 'install-exec-am'.
/usr/bin/mkdir -p '/mnt/lfs/usr/share/man/man1'
/usr/bin/install -c -m 644 'patch.man' '/mnt/lfs/usr/share/man/man1/patch.1'
make[3]: Leaving directory '/mnt/lfs/sources/patch-2.8'
make[2]: Leaving directory '/mnt/lfs/sources/patch-2.8'
make[1]: Leaving directory '/mnt/lfs/sources/patch-2.8'
lfs:/mnt/lfs/sources/patch-2.8$

```

-Capítulo 6.14 hecho a las 10:45.

- Instalación del paquete Sed-4.9, que contiene un editor de stream.
- Comandos principales ejecutados: configure, make, make DESTDIR=\$LFS install.

```

make[5]: Nothing to be done for 'install-exec-am'.
make[5]: Nothing to be done for 'install-data-am'.
make[5]: Leaving directory '/mnt/lfs/sources/sed-4.9/gnulib-tests'
make[4]: Leaving directory '/mnt/lfs/sources/sed-4.9/gnulib-tests'
make[3]: Leaving directory '/mnt/lfs/sources/sed-4.9/gnulib-tests'
make[2]: Leaving directory '/mnt/lfs/sources/sed-4.9/gnulib-tests'
make[1]: Leaving directory '/mnt/lfs/sources/sed-4.9'
lfs:/mnt/lfs/sources/sed-4.9$

```

-Capítulo 6.15 hecho a las 11:05.

- Instalación del paquete Tar-1.35, que proporciona la capacidad de crear archivos tar, así como de realizar varios otros tipos de manipulación de archivos. Tar puede utilizarse sobre archivos creados previamente para extraer archivos, almacenar archivos adicionales o actualizar o listar archivos que ya estaban almacenados.
- Comandos principales ejecutados: configure, make, make DESTDIR=\$LFS install.

```

make[1]: Leaving directory '/mnt/lfs/sources/tar-1.35/po'
Making install in tests
make[1]: Entering directory '/mnt/lfs/sources/tar-1.35/tests'
make[2]: Entering directory '/mnt/lfs/sources/tar-1.35/tests'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/mnt/lfs/sources/tar-1.35/tests'
make[1]: Leaving directory '/mnt/lfs/sources/tar-1.35/tests'
make[1]: Entering directory '/mnt/lfs/sources/tar-1.35'
make[2]: Entering directory '/mnt/lfs/sources/tar-1.35'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/mnt/lfs/sources/tar-1.35'
make[1]: Leaving directory '/mnt/lfs/sources/tar-1.35'
lfs:/mnt/lfs/sources/tar-1.35$

```

-Capítulo 6.16 hecho a las 11:45.

- Instalación del paquete Xz-5.8.1, que contiene programas para comprimir y descomprimir archivos. Proporciona compatibilidad con los formatos de compresión lzma y el más reciente xz. Al comprimir archivos de texto con xz se obtiene un mejor porcentaje de compresión que con los comandos tradicionales gzip o bzip2.
- Comandos principales ejecutados: configure, make, make DESTDIR=\$LFS install, rm.

```

make[1]: Leaving directory '/mnt/lfs/sources/xz-5.8.1/po'
Making install in tests
make[1]: Entering directory '/mnt/lfs/sources/xz-5.8.1/tests'
make[2]: Entering directory '/mnt/lfs/sources/xz-5.8.1/tests'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/mnt/lfs/sources/xz-5.8.1/tests'
make[1]: Leaving directory '/mnt/lfs/sources/xz-5.8.1/tests'
make[1]: Entering directory '/mnt/lfs/sources/xz-5.8.1'
make[2]: Entering directory '/mnt/lfs/sources/xz-5.8.1'
make[2]: Nothing to be done for 'install-exec-am'.
/usr/bin/mkdir -p '/mnt/lfs/usr/share/doc/xz-5.8.1'
/usr/bin/install -c -m 644 AUTHORS COPYING.README BSD COPYING.GPLv2 NEWS README THANKS doc/faq.txt doc/history.txt doc/xz-file-format.txt doc/lzma-file-format.txt '/mnt/lfs/usr/share/doc/xz-5.8.1'
/usr/bin/mkdir -p '/mnt/lfs/usr/share/doc/xz-5.8.1/examples'
/usr/bin/install -c -m 644 doc/examples/00_README.txt doc/examples/01_compress_easy.c doc/examples/02_decompress.c doc/examples/03_compress_custom.c doc/examples/04_compress_easy_mt.c doc/examples/11_file_info.c doc/examples/Makefile '/mnt/lfs/usr/share/doc/xz-5.8.1/examples'
make[2]: Leaving directory '/mnt/lfs/sources/xz-5.8.1'
make[1]: Leaving directory '/mnt/lfs/sources/xz-5.8.1'
lfs:/mnt/lfs/sources/xz-5.8.1$ rm -v $LFS/usr/lib/liblzma.la
removed '/mnt/lfs/usr/lib/liblzma.la'
lfs:/mnt/lfs/sources/xz-5.8.1$

```

-Capítulo 6.17 hecho a las 13:00.

- Instalación del paquete Binutils-2.45 (Pass 2), que contiene un enlazador, un ensamblador y otras herramientas para manejar archivos objeto.
- Comandos principales ejecutados: sed, mkdir, cd, configure, make, make DESTDIR=\$LFS install, rm.

Obs: en este capítulo me di cuenta de que se supone que tenía que borrar las carpetas creadas por el tar...

```

make[3]: Entering directory '/mnt/lfs/sources/binutils-2.45/build/libiberty/testsuite'
make[3]: Nothing to be done for 'install'.
make[3]: Leaving directory '/mnt/lfs/sources/binutils-2.45/build/libiberty/testsuite'
make[2]: Leaving directory '/mnt/lfs/sources/binutils-2.45/build/libiberty'
make[1]: Nothing to be done for 'install-target'.
make[1]: Leaving directory '/mnt/lfs/sources/binutils-2.45/build'
lfs:/mnt/lfs/sources/binutils-2.45/build$ rm -v $LFS/usr/lib/libbfd,ctf,ctf-nobfd,opcodes,sframe}.{a,la}
removed '/mnt/lfs/usr/lib/libbfd.a'
removed '/mnt/lfs/usr/lib/libbfd.la'
removed '/mnt/lfs/usr/lib/libctf.a'
removed '/mnt/lfs/usr/lib/libctf.la'
removed '/mnt/lfs/usr/lib/libctf-nobfd.a'
removed '/mnt/lfs/usr/lib/libctf-nobfd.la'
removed '/mnt/lfs/usr/lib/libopcodes.a'
removed '/mnt/lfs/usr/lib/libopcodes.la'
removed '/mnt/lfs/usr/lib/libsframe.a'
removed '/mnt/lfs/usr/lib/libframe.la'
lfs:/mnt/lfs/sources/binutils-2.45/build$

```

-Capítulo 6.18 hecho a las 15:15 (Empezado a las 11:30...).

- Instalación del paquete GCC-15.2.0 (Pass 2), que contiene la colección de compiladores GNU, que incluye los compiladores de C y C++.
- Comandos principales ejecutados: tar, mv, uname, sed, mkdir, cd, configure, make, make DESTDIR=\$LFS install, ln.

## Capítulo 7

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-Capítulo 7.2 hecho a las 11:05.

- Cambio de propietario.
- Comandos principales ejecutados: chown, uname.

-Capítulo 7.3 hecho a las 11:10.

- Preparación de Sistemas de Archivos Virtuales del Kernel.
- Montaje de los Sistemas de Archivos Virtuales del Kernel.
- Comandos principales ejecutados: mkdir, mount, install.

-Capítulo 7.4 (Entering the Chroot Environment) hecho a las 11:15.

- Entrada al Entorno Chroot.
- Comandos principales ejecutados: chroot.

```
[root@localhost ~]# chown --from lfs -R root:root $LFS/{usr,var,etc,tools}
[root@localhost ~]# case $(uname -m) in
> x86_64) chown -from lfs -R root:root $LFS/lib64 ;;
> esac
chown: opción inválida -- 'r'
Pruebe 'chown --help' para más información.
[root@localhost ~]# case $(uname -m) in x86_64) chown --from lfs -R root:root $LFS/lib64 ;; esac
[root@localhost ~]# mkdir -pv $LFS/{dev,proc,sys,run}
mkdir: se ha creado el directorio '/mnt/lfs/dev'
mkdir: se ha creado el directorio '/mnt/lfs/proc'
mkdir: se ha creado el directorio '/mnt/lfs/sys'
mkdir: se ha creado el directorio '/mnt/lfs/run'
[root@localhost ~]# mount -v --bind /dev $LFS/dev
mount: /dev bound on /mnt/lfs/dev.
[root@localhost ~]# mount -vt devpts devpts -o gid=5,mode=0620 $LFS/dev/pts
mount: devpts mounted on /mnt/lfs/dev/pts.
[root@localhost ~]# mount -vt proc proc $LFS/proc
mount: proc mounted on /mnt/lfs/proc.
[root@localhost ~]# mount -vt sysfs sysfs $LFS/sys
mount: sysfs mounted on /mnt/lfs/sys.
[root@localhost ~]# mount -vt tmpfs tmpfs $LFS/run
mount: tmpfs mounted on /mnt/lfs/run.
[root@localhost ~]# if [ -h $LFS/dev/shm ]; then
> install -v -d -m 1777 $LFS$(realpath /dev/shm)
> else
> mount -vt tmpfs -o nosuid,nodev tmpfs $LFS/dev/shm
> fi
mount: tmpfs mounted on /mnt/lfs/dev/shm.
[root@localhost ~]# chroot "$LFS" /usr/bin/env -i \
> HOME=/root \
> TERM="$TERM" \
> PS1='(lfs chroot) \u:\w\$ ' \
> PATH=/usr/bin:/usr/sbin \
> MAKEFLAGS="-j$(nproc)" \
> TESTSUITEFLAGS="-j$(nproc)" \
> /bin/bash --login
(lfs chroot) I have no name!:/#
```

-Capítulo 7.5 hecho a las 11:20.

- Creación de directorios.
- Comandos principales ejecutados: mkdir, ln, install.

-Capítulo 7.6 hecho a las 11:40.

- Creación de Archivos Esenciales y Symlinks.
- Comandos principales ejecutados: ln, cat, echo, install, exec, touch, chgrp, chmod.

-Capítulo 7.7 hecho a las 11:50.

- Instalación del paquete Gettext-0.26, que contiene utilidades para internacionalización y localización. Estas permiten que los programas se compilen con NLS (Soporte de Idioma Nativo), habilitándolos para mostrar mensajes en el idioma nativo del usuario.
- Comandos principales ejecutados: tar, configure, make, cp.

-Capítulo 7.8 hecho a las 12:50.

- Instalación de paquete Bison-3.8.2, que contiene un generador de parsers.
- Comandos principales ejecutados: tar, configure, make, make install.

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-Capítulo 7.9 hecho a las 23:50.

- Instalación del paquete Perl-5.42.0, que contiene the Practical Extraction y Report Language.
- Comandos principales ejecutados: tar, configure, make, make install.

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-Capítulo 7.10 hecho a las 0:40.

- Instalación del paquete Python-3.13.7, que contiene el entorno de desarrollo de Python. Es útil para programación orientada a objetos, escritura de scripts, creación de prototipos de programas grandes y desarrollo de aplicaciones completas. Python es un lenguaje de programación interpretado.
- Comandos principales ejecutados: tar, configure, make, make install.

-Capítulo 7.11 hecho a las 0:47.

- Instalación del paquete Texinfo-7.2, que contiene programas para lectura, escritura y conversión de info pages.
- Comandos principales ejecutados: tar, configure, make, make install.

-Capítulo 7.12 hecho a las 01:12.

- Instalación del paquete Util-linux-2.41.1, que contiene misceláneos programas utilitarios.
- Comandos principales ejecutados: tar, mkdir, configure, make, make install.



-Capítulo 7.13 hecho a las 01:15.

- Limpieza.
- Comandos principales ejecutados: rm, find.

## Capítulo 8

06/12/2025

-Capítulo 8.3 hecho a las 9:50

-Capítulo 8.4 hecho a las 9:55

```
INSTALL      /usr/share/man/man3/intro.3
INSTALL      /usr/share/man/man3attr/intro.3attr
INSTALL      /usr/share/man/man4/intro.4
INSTALL      /usr/share/man/man5/intro.5
INSTALL      /usr/share/man/man6/intro.6
INSTALL      /usr/share/man/man7/intro.7
INSTALL      /usr/share/man/man8/intro.8
INSTALL      /usr/share/man/man2const/EPIOCGPARAMS.2const
INSTALL      /usr/share/man/man2type/open_how.2type
INSTALL      /usr/share/man/man3const/EOF.3const
INSTALL      /usr/share/man/man3head/printf.h.3head
INSTALL      /usr/share/man/man3head/sysexits.h.3head
INSTALL      /usr/share/man/man3type/aio.3type
(lfs chroot) root:/sources/man-pages-6.15# cd ..
(lfs chroot) root:/sources# rm -rf man-pages-6.15
(lfs chroot) root:/sources# ls
Python-3.13.7.tar.xz          expect-5.45.4-gcc15-1.patch.1  kbd-2.8.0.tar.xz
XML-Parser-2.47.tar.gz       expect5.45.4.tar.gz            kmod-34.2.tar.xz
acl-2.3.2.tar.xz             file-5.46.tar.gz               less-679.tar.gz
attr-2.5.2.tar.gz            findutils-4.10.0.tar.xz       lfs-bootscripts-2025
autoconf-2.72.tar.xz         flex-2.6.4.tar.gz             libcap-2.76.tar.xz
automake-1.18.1.tar.xz       flit_core-3.12.0.tar.gz       libffi-3.5.2.tar.gz
bash-5.3.tar.gz              gawk-5.3.2.tar.xz             libpipeline-1.5.8.ta
bc-7.0.3.tar.xz              gcc-15.2.0.tar.xz             libtool-2.5.4.tar.xz
binutils-2.45.tar.xz         gdbm-1.26.tar.gz              libxcrypt-4.4.38.tar
bison-3.8.2.tar.xz          gettext-0.26.tar.xz          linux-6.16.1.tar.xz
bzip2-1.0.8-install_docs-1.patch  glibc-2.42-fhs-1.patch      lz4-1.10.0.tar.gz
bzip2-1.0.8-install_docs-1.patch.1  glibc-2.42-fhs-1.patch.1    m4-1.4.20.tar.xz
bzip2-1.0.8.tar.gz           glibc-2.42.tar.xz            make-4.4.1.tar.gz
coreutils-9.7-i18n-1.patch      gmp-6.3.0.tar.xz             man-db-2.13.1.tar.xz
coreutils-9.7-upstream_fix-1.patch  gperf-3.3.tar.gz             man-pages-6.15.tar.x
coreutils-9.7-upstream_fix-1.patch.1  grep-3.12.tar.gz            markupsafe-3.0.2.tar
coreutils-9.7.tar.xz           groff-1.23.0.tar.gz          md5sums
dbus-1.16.2.tar.xz            grub-2.12.tar.xz              meson-1.8.3.tar.gz
dejagmu-1.6.3.tar.gz           gzip-1.14.tar.xz             mpc-1.3.1.tar.gz
diffutils-3.12.tar.xz         iana-etc-20250807.tar.gz     mpfr-4.2.2.tar.xz
download                     inetutils-2.6.tar.xz         ncurses-6.5-20250809
e2fsprogs-1.47.3.tar.gz        intltool-0.51.0.tar.gz       ninja-1.13.1.tar.gz
elfutils-0.193.tar.bz2         iproute2-6.16.0.tar.xz      openssl-3.5.4.tar.gz
expat-2.7.3.tar.xz            jinja2-3.1.6.tar.gz          packaging-25.0.tar.g
expect-5.45.4-gcc15-1.patch     kbd-2.8.0-backspace-1.patch  patch-2.8.tar.xz
expect-5.45.4-gcc15-1.patch.1  kbd-2.8.0-backspace-1.patch.1  perl-5.42.0.tar.xz
(lfs chroot) root:/sources# tar -xzf iana-etc-20250807.tar.gz
(lfs chroot) root:/sources# cd iana-etc-20250807
(lfs chroot) root:/sources/iana-etc-20250807# cp services protocols /etc
(lfs chroot) root:/sources/iana-etc-20250807# cd ..
(lfs chroot) root:/sources# rm -rf iana-etc-20250807
```

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-Capítulo 8.5.1 hecho a las 16:05

```

(lfs chroot) root:/sources# tar -xf glibc-2.42.tar.xz
(lfs chroot) root:/sources# cd glibc-2.42
(lfs chroot) root:/sources/glibc-2.42# patch -Np1 -i ../glib-2.42-fhs-1.patch
patch: **** Can't open patch file ../glib-2.42-fhs-1.patch : No such file or directory
(lfs chroot) root:/sources/glibc-2.42# patch -Np1 -i ../glibc-2.42-fhs-1.patch
patching file Makeconfig
Hunk #1 succeeded at 262 (offset 12 lines).
patching file nsd/nsd.h
Hunk #1 succeeded at 160 (offset 48 lines).
patching file nss/db-Makefile
Hunk #1 succeeded at 21 (offset -1 lines).
patching file sysdeps/generic/paths.h
patching file sysdeps/unix/sysv/linux/paths.h
(lfs chroot) root:/sources/glibc-2.42# sed -e '/unistd.h/i #include <string.h>' \
> -e '/libc_rwlock_init/c\
> __libc_rwlock_define_initialized(, reset_lock);\
> memcpy(&lock, &reset_lock, sizeof(lock));' \
> -i stdlib/abort.c
(lfs chroot) root:/sources/glibc-2.42# mkdir -v build
mkdir: created directory 'build'
(lfs chroot) root:/sources/glibc-2.42# cd build
(lfs chroot) root:/sources/glibc-2.42/build# echo "rootsbindir=/usr/sbin" > configparms
(lfs chroot) root:/sources/glibc-2.42/build# ../configure --prefix=/usr \
> --disable-werror \
> --disable-nsd \
> libc_cv_slibdir=/usr/lib \
> --enable-stack-protector=strong \
> --enable-kernel=5.4

```

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-Capítulo 8.5.2 y 8.5.3 hecho a las 23:59

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-Capítulo 8.6 hecho a las 00:07

-Capítulo 8.7 hecho a las 00:14

-Capítulo 8.8 hecho a las 00:22

-Capítulo 8.9 hecho a las 00:30

-Capítulo 8.10 hecho a las 00:37

-Capítulo 8.11 hecho a las 00:45

-Capítulo 8.12 hecho a las 01:10

-Capítulo 8.13 hecho a las 01:30

-Capítulo 8.14 hecho a las 01:33

-Capítulo 8.15 hecho a las 01:42

```

(lfs chroot) root:/sources/flex-2.6.4# ln -sv flex /usr/bin/lex
'/usr/bin/lex' -> 'flex'
(lfs chroot) root:/sources/flex-2.6.4# ln -sv flex.1 /usr/share/man/man1/lex.1
'/usr/share/man/man1/lex.1' -> 'flex.1'
(lfs chroot) root:/sources/flex-2.6.4# cd ..
(lfs chroot) root:/sources# rm -rf flex-2.6.4
(lfs chroot) root:/sources#

```

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-Capitulo 8.16 hecho a las 00:10

```
Tests ended at Sun Oct 19 07:03:06 -03 2025
all.tcl:      Total  47105  Passed 43684  Skipped 3421  Failed  0
Sourced 152 Test Files.
Number of tests skipped for each constraint:
    9  !ieeeFloatingPoint
    1  !singleTestInterp notWsl
    1  Unix
    3  asyncPipeChan
   76  bigEndian
    5  bug-3057639
   49  dde
    1  dontCopyLinks
   62  emptyTest
    2  extensive
   58  fullutf
    2  hasIsoLocale
    1  knownBadTest
   46  knownBug
  100  localeRegexp
    9  localhost_v6
   47  longIs32bit
   14  macosxFileAttr
   85  memory
   42  nonPortable
  258  notRoot
    2  notValgrind
    4  nt
    1  pairsTo4bytes
    2  pre388
 1996  serverNeeded
    1  testWithLimit
    1  testexprparser && !ieeeFloatingPoint
    1  testwinclock
    2  unthreaded
   17  utf16
   21  utf32
    2  wideBiggerThanInt
  494  win
    1  winLessThan10
    3  winVista
    2  xdev
(lfs chroot) root:/sources/tcl8.6.16/unix# make install
```

-Capitulo 8.17 hecho a las 00:23

```
(lfs chroot) root:/sources/expect5.45.4# ln -srf expect5.45.4/libexpect5.45.4.so /usr/lib
'/usr/lib/libexpect5.45.4.so' -> 'expect5.45.4/libexpect5.45.4.so'
(lfs chroot) root:/sources/expect5.45.4# cd ..
(lfs chroot) root:/sources# rm -rf expect5.45.4
```

-Capitulo 8.18 hecho a las 00:40

-Capitulo 8.19 hecho a las 00:56

```
-----
Libraries have been installed in:
  /usr/lib
```

If you ever happen to want to link against installed libraries in a given directory, LIBDIR, you must either use libtool, and specify the full pathname of the library, or use the '-LLIBDIR' flag during linking and do at least one of the following:

- add LIBDIR to the 'LD\_LIBRARY\_PATH' environment variable during execution
- add LIBDIR to the 'LD\_RUN\_PATH' environment variable during linking
- use the '-Wl,-rpath -Wl,LIBDIR' linker flag
- have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for more information, such as the ld(1) and ld.so(8) manual pages.

```
-----
/usr/bin/mkdir -p '/usr/bin'
/bin/sh ./libtool --mode=install /usr/bin/install -c pkgconf bomtool '/usr/bin'
libtool: install: /usr/bin/install -c .libs/pkgconf /usr/bin/pkgconf
libtool: install: /usr/bin/install -c .libs/bomtool /usr/bin/bomtool
make[1]: Leaving directory '/sources/pkgconf-2.5.1'
(lfs chroot) root:/sources/pkgconf-2.5.1# ln -sv pkgconf /usr/bin/pkg-config
'/usr/bin/pkg-config' -> 'pkgconf'
(lfs chroot) root:/sources/pkgconf-2.5.1# ln -sv pkgconf.1 /usr/share/man/man1/pkg-config.1
'/usr/share/man/man1/pkg-config.1' -> 'pkgconf.1'
(lfs chroot) root:/sources/pkgconf-2.5.1#
```

-Capitulo 8.20 hecho a las 01:40

```
(lfs chroot) root:/sources/binutils-2.45/build# rm -rfv /usr/lib/libbfd,ctf,ctf-nobfd,gprofng,opcodes,sframe}.a \
> /usr/share/doc/gprofng/
removed '/usr/lib/libbfd.a'
removed '/usr/lib/libctf.a'
removed '/usr/lib/libctf-nobfd.a'
removed '/usr/lib/libgprofng.a'
removed '/usr/lib/libopcodes.a'
removed '/usr/lib/libsfame.a'
removed '/usr/share/doc/gprofng/examples.tar.gz'
removed directory '/usr/share/doc/gprofng/'
(lfs chroot) root:/sources/binutils-2.45/build#
```

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-Capitulo 8.21 hecho a las 19:29

```
(lfs chroot) root:/sources/gmp-6.3.0# awk '/# PASS:/{total+=1} ; END{print total}' gmp-check-log
199
```

-Capitulo 8.22 hecho a las 19:45

```
# TOTAL: 198
# PASS: 198
# SKIP: 0
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====
```

-Capitulo 8.23 hecho a las 19:54

```

=====
Testsuite summary for mpc 1.3.1
=====
# TOTAL: 74
# PASS: 74
# SKIP: 0
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====

```

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-Capitulo 8.24 (attr-2.5.2) hecho a las 01:54

```

(lfs chroot) root:/sources/attr-2.5.2# getfattr -d /tmp/prueba_attr
getfattr: /tmp/prueba_attr: No such file or directory
(lfs chroot) root:/sources/attr-2.5.2# touch /tmp/prueba_attr
(lfs chroot) root:/sources/attr-2.5.2# setfattr -n user.test -v hola /tmp/prueba_attr
(lfs chroot) root:/sources/attr-2.5.2# getfattr /tmp/prueba_attr
getfattr: Removing leading '/' from absolute path names
# file: tmp/prueba_attr
user.test

(lfs chroot) root:/sources/attr-2.5.2# getfattr -d /tmp/prueba_attr
getfattr: Removing leading '/' from absolute path names
# file: tmp/prueba_attr
user.test="hola"

(lfs chroot) root:/sources/attr-2.5.2#

```

-Capitulo 8.25 (acl-2.3.2) hecho a las 02:11

```

(lfs chroot) root:/sources/acl-2.3.2# setfacl -m u:root:rwx /root
(lfs chroot) root:/sources/acl-2.3.2# getfacl /root | head
getfacl: Removing leading '/' from absolute path names
# file: root
# owner: root
# group: root
user::rwx
user:root:rwx
group::r-x
mask::rwx
other::---

```

-Capitulo 8.26 hecho a las 02:33

-Capitulo 8.27 hecho a las 02:50

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-Capitulo 8.28 (shadow-4.18) hecho a las 17:00

```
(lfs chroot) root:/sources/shadow-4.18.0# pwconv
(lfs chroot) root:/sources/shadow-4.18.0# grpconv
(lfs chroot) root:/sources/shadow-4.18.0# mkdir -p /etc/default
(lfs chroot) root:/sources/shadow-4.18.0# useradd -D --gid 999
useradd: unrecognized option '--gid'
Usage: useradd [options] LOGIN
        useradd -D
        useradd -D [options]

Options:
        --badname          do not check for bad names
-b, --base-dir BASE_DIR   base directory for the home directory of the
                           new account
        --btrfs-subvolume-home use BTRFS subvolume for home directory
-c, --comment COMMENT     GECOS field of the new account
-d, --home-dir HOME_DIR   home directory of the new account
-D, --defaults            print or change default useradd configuration
-e, --expiredate EXPIRE_DATE expiration date of the new account
-f, --inactive INACTIVE   password inactivity period of the new account
-F, --add-subids-for-system add entries to subuid even when adding a system user
-g, --gid GROUP           name or ID of the primary group of the new
                           account
-G, --groups GROUPS       list of supplementary groups of the new
                           account
-h, --help               display this help message and exit
-k, --skel SKEL_DIR      use this alternative skeleton directory
-K, --key KEY=VALUE       override /etc/login.defs defaults
-m, --create-home         create the user's home directory
-M, --no-create-home      do not create the user's home directory
-N, --no-user-group       do not create a group with the same name as
                           the user
-o, --non-unique          allow to create users with duplicate
                           (non-unique) UID
-p, --password PASSWORD   encrypted password of the new account
-r, --system             create a system account
-R, --root CHROOT_DIR     directory to chroot into
-P, --prefix PREFIX_DIR   prefix directory where are located the /etc/* files
-s, --shell SHELL         login shell of the new account
-u, --uid UID            user ID of the new account
-U, --user-group          create a group with the same name as the user
```

```
(lfs chroot) root:/sources/shadow-4.18.0# passwd root
Changing password for root
Enter the new password (minimum of 5 characters)
Please use a combination of upper and lower case letters and numbers.
The password for root is unchanged.
(lfs chroot) root:/sources/shadow-4.18.0#
```

```
(lfs chroot) root:/sources/shadow-4.18.0# make -C man install-man
make: Entering directory '/sources/shadow-4.18.0/man'
/usr/bin/mkdir -p '/usr/share/man/man1'
/usr/bin/mkdir -p '/usr/share/man/man3'
/usr/bin/install -c -m 644 man1/chage.1 man1/chfn.1 man1/chsh.1 man1/expiry.1 man1/gpasswd.1 man1/login.1 man1/newgrp.1
getsubids.1 man1/newgidmap.1 man1/newuidmap.1 '/usr/share/man/man1'
/usr/bin/install -c -m 644 man3/shadow.3 '/usr/share/man/man3'
/usr/bin/mkdir -p '/usr/share/man/man8'
/usr/bin/mkdir -p '/usr/share/man/man5'
/usr/bin/install -c -m 644 man5/faillog.5 man5/gshadow.5 man5/login.defs.5 man5/shadow.5 man5/limits.5 man5/login.access
ubgid.5 man5/subuid.5 '/usr/share/man/man5'
/usr/bin/install -c -m 644 man8/chgpasswd.8 man8/chpasswd.8 man8/faillog.8 man8/groupadd.8 man8/groupdel.8 man8/groupme
rpconv.8 man8/grpconv.8 man8/logoutd.8 man8/newusers.8 man8/nologin.8 man8/pwck.8 man8/pwconv.8 man8/pwunconv.8 man8/u
an8/vigr.8 man8/vipw.8 '/usr/share/man/man8'
make: Leaving directory '/sources/shadow-4.18.0/man'
(lfs chroot) root:/sources/shadow-4.18.0# pwconv
(lfs chroot) root:/sources/shadow-4.18.0# grpconv
(lfs chroot) root:/sources/shadow-4.18.0# mkdir -p /etc/default
(lfs chroot) root:/sources/shadow-4.18.0# useradd -D --gid 999
(lfs chroot) root:/sources/shadow-4.18.0# passwd root
Changing password for root
Enter the new password (minimum of 5 characters)
Please use a combination of upper and lower case letters and numbers.
New password:
Re-enter new password:
passwd: password changed.
(lfs chroot) root:/sources/shadow-4.18.0#
(lfs chroot) root:/sources/shadow-4.18.0#
```

-Capítulo 8.29 (gcc-15.2.0) hecho a las 16:00

Obs: 23 horas...

```
make[4]: Leaving directory '/sources/gcc-15.2.0/build/x86_64-pc-linux-gnu/libatomic'
make[3]: Leaving directory '/sources/gcc-15.2.0/build/x86_64-pc-linux-gnu/libatomic'
make[2]: Leaving directory '/sources/gcc-15.2.0/build/x86_64-pc-linux-gnu/libatomic'
make[1]: Leaving directory '/sources/gcc-15.2.0/build'

real    237m0.933s
user    219m19.145s
sys      8m17.330s
(lfs chroot) root:/sources/gcc-15.2.0/build#
```

```
=====
Testsuite summary for package-unused version-unused
=====
```

```
# TOTAL: 38
# PASS:  38
# SKIP:  0
# XFAIL: 0
# FAIL:  0
# XPASS: 0
# ERROR: 0
=====
```

```
=== g++ Summary ===
# of expected passes      248517
# of expected failures    2282
# of unsupported tests     2049
/sources/gcc-15.2.0/build/gcc/xg++ version 15.2.0 (GCC)
```

```
=== gcc tests ===
--
=== gcc Summary ===
# of expected passes      211470
# of unexpected failures   4
# of expected failures    1476
# of unsupported tests     3784
/sources/gcc-15.2.0/build/gcc/xgcc version 15.2.0 (GCC)
```

```
=== libatomic Summary ===
# of expected passes      54
=== libgomp tests ===
```

Running target unix

```
=== libgomp Summary ===
# of expected passes      5988
# of expected failures    46
# of unsupported tests     398
=== libitm tests ===
```

```
=== libitm Summary ===
# of expected passes      44
# of expected failures    3
# of unsupported tests     1
```

```

=== libstdc++ Summary ===

# of expected passes      19099
# of expected failures    131
# of unsupported tests     392

Compiler version: 15.2.0 (GCC)
Platform: x86_64-pc-linux-gnu
(lfs chroot) root:/sources/gcc-15.2.0/build#

```

```

=== gcc Summary ===

# of expected passes      102492
# of unexpected failures   4
# of expected failures    805
# of unsupported tests     1621
/sources/gcc-15.2.0/build/gcc/xgcc version 15.2.0 (GCC)

```

```

(lfs chroot) root:/sources/gcc-15.2.0/build# ln -svr /usr/bin/cpp /usr/lib
'/usr/lib/cpp' -> './bin/cpp'
(lfs chroot) root:/sources/gcc-15.2.0/build# ln -sv gcc.1 /usr/share/man/man1/cc.1
'/usr/share/man/man1/cc.1' -> 'gcc.1'
(lfs chroot) root:/sources/gcc-15.2.0/build# ln -sfv ../../libexec/gcc/${gcc -dumpmachine}/15.2.0/liblto_plugin.so \
> /usr/lib/bfd-plugins/
'/usr/lib/bfd-plugins/liblto_plugin.so' -> '../../libexec/gcc/x86_64-pc-linux-gnu/15.2.0/liblto_plugin.so'
(lfs chroot) root:/sources/gcc-15.2.0/build# echo 'int main(){ }' | cc -x c - -v -Wl,--verbose &> dummy.log
(lfs chroot) root:/sources/gcc-15.2.0/build# readelf -l a.out | grep ': /lib'
[Requesting program interpreter: /lib64/ld-linux-x86-64.so.2]
(lfs chroot) root:/sources/gcc-15.2.0/build# grep -E -o '/usr/lib.*S?crt[1in].*succeeded' dummy.log
/usr/lib/gcc/x86_64-pc-linux-gnu/15.2.0/../../../../lib/Scrt1.o succeeded
/usr/lib/gcc/x86_64-pc-linux-gnu/15.2.0/../../../../lib/crti.o succeeded
/usr/lib/gcc/x86_64-pc-linux-gnu/15.2.0/../../../../lib/crtn.o succeeded
(lfs chroot) root:/sources/gcc-15.2.0/build# grep -B4 '^ /usr/include' dummy.log
#include <...> search starts here:
 /usr/lib/gcc/x86_64-pc-linux-gnu/15.2.0/include
 /usr/local/include
 /usr/lib/gcc/x86_64-pc-linux-gnu/15.2.0/include-fixed
 /usr/include
(lfs chroot) root:/sources/gcc-15.2.0/build# grep 'SEARCH.*usr/lib' dummy.log | sed 's|; |\n|g'
SEARCH_DIR("/usr/x86_64-pc-linux-gnu/lib64")
SEARCH_DIR("/usr/local/lib64")
SEARCH_DIR("/lib64")
SEARCH_DIR("/usr/lib64")
SEARCH_DIR("/usr/x86_64-pc-linux-gnu/lib")
SEARCH_DIR("/usr/local/lib")
SEARCH_DIR("/lib")
SEARCH_DIR("/usr/lib");
(lfs chroot) root:/sources/gcc-15.2.0/build# grep "/lib.*libc.so.6 " dummy.log
attempt to open /usr/lib/libc.so.6 succeeded
(lfs chroot) root:/sources/gcc-15.2.0/build# grep found dummy.log
found ld-linux-x86-64.so.2 at /usr/lib/ld-linux-x86-64.so.2
(lfs chroot) root:/sources/gcc-15.2.0/build# rm -v a.out dummy.log
removed 'a.out'
removed 'dummy.log'
(lfs chroot) root:/sources/gcc-15.2.0/build# mkdir -pv /usr/share/gdb/auto-load/usr/lib
mkdir: created directory '/usr/share/gdb'
mkdir: created directory '/usr/share/gdb/auto-load'
mkdir: created directory '/usr/share/gdb/auto-load/usr'
mkdir: created directory '/usr/share/gdb/auto-load/usr/lib'
(lfs chroot) root:/sources/gcc-15.2.0/build# mv -v /usr/lib/*gdb.py /usr/share/gdb/auto-load/usr/lib
renamed '/usr/lib/libstdc++.so.6.0.34-gdb.py' -> '/usr/share/gdb/auto-load/usr/lib/libstdc++.so.6.0.34-gdb.py'
(lfs chroot) root:/sources/gcc-15.2.0/build#

```

-Capítulo 8.30 (ncurses) hecho a las 18:00



```

(rlfs chroot) root:/sources/ncurses-6.5-20250809/testst ls
ls
lsakefile combine.c dots_curses.c form_driver.w.c listused.sh programs test_endwin testaddch
lsakefile.in configure dots_curses.c gdc lrtest railroad test_endwin.c testaddch.c
README configure.in dots_mcurc gdc.6 lrtest.c railroad.c test_get_wstr testcurs.c
lslocal.mf demo_allkeys dots_mcurc.gdc.c make_tar.sh rain test_get_wstr.c testcurs.c
lsback_ground.c demo_allkeys.c dots_termcap.hanoi.c merge_menus.xbm rail test_getstr testcurs.c
lsback_ground.c demo_defkey dots_termcap.c hanoi.c mk-test awk redrau test_instr testcurs.c
lsback_ground.c demo_forms dots_xcurses.c hashtest modules savescreen test_instr.c tput-colorcube
lsblue.c demo_forms.c dump_window.c inch_wide move_field savescreen.c test_instr.c tput-initc
lsblue.c demo_forms.txt dump_window.h inch_wide.c move_field.c savescreen.sh test_instr.c view
lsbs bs.c demo_keyok dup_field inchs movewindow sp_tinfo.c test_mouse view.c
lsbs.c demo_keyok.c dup_field.c inchs.c movewindow.c sp_tinfo.c test_mouse.c widechars-utf8-tabs.txt
lsbs.c demo_menus.c echochar.h lns_wide ncurses.tcl test_opaque widechars-utf8.txt
lsbulgarian-utf8-tabs.txt demo_menus.c echowchar.c lns_wide.c tcl test_opaque.c widechars.h
lsbulgarian-utf8.txt demo_new_pair edit_field.c insdell ncurses.tst.hin test_setupterm work
lsbulgarian-utf8.txt demo_new_pair.c edit_field.c insdell.c newdemo test_priv.h test_setupterm.c work.c
lsbulgarian-utf8.txt demo_panels.c escheknort.xbm inserts newdemo.c test_add_wchstr test_sgr xms
lsbulgarian-utf8.txt demo_panels.c extended_color inserts.c newdemo.c test_add_wchstr.c test_sgr.c xms.c
lschgat chgat.c demo_tabs key_names package padview test_addchstr test_termattrs xterm-16color.dat
lschgat.c demo_tabs.c filter key_names.c package padview.c test_addchstr.c test_termattrs.c xterm-256color.dat
lschgat.c demo_termcap filter.c keynames pair_content test_addstr test_tparm xterm-88color.dat
lschgat.c demo_termcap.c filter.c keynames.c pair_content.c test_addstr.c test_tparm.c xterm-color_40x40.xpm
lschgat.c demo_termcap.c firework.c knight test_addstr test_unget_wch.c
lschgat.c demo_terminfo.c firstlast.c knight.c picsmap.c test_arrays test_vid_puts
lschgat.c demo_termname.c ditto firstlast.c lnsdata.h picsmap.c test_arrays.c test_vid_puts.c
lschgat.c demo_termname.c ditto.c foldkeys linux-color.dat picsmap.h test_arrays.c test_vid_puts.c
lschgat.c demo_termname.c dots foldkeys.c list_keys popup_msg.c test_delwin test_vidputs
lscombine combine.c dots.c form_driver.w list_keys.c popup_msg.h test_delwin test_vidputs

```

```

TOWERS OF HANOI
SJR 1990

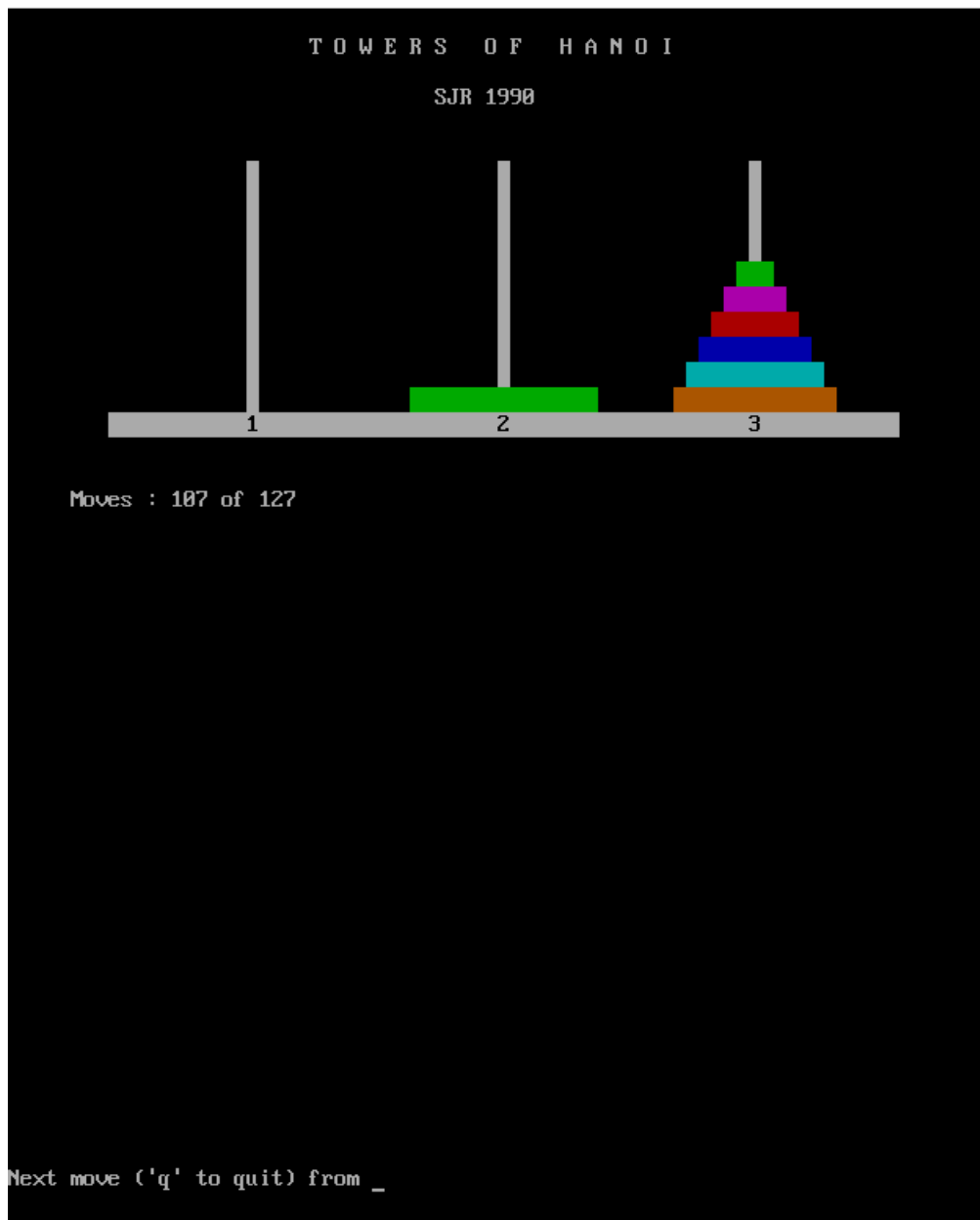
```

```

Moves : 97 of 127

```

Next move ('q' to quit) from \_



-Capitulo 8.31 (sed) hecho a las 18:05

```

=====
Testsuite summary for GNU sed 4.9
=====
# TOTAL: 192
# PASS: 171
# SKIP: 21
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====
make[6]: Leaving directory '/sources/sed-4.9/gnulib-tests'
make[5]: Leaving directory '/sources/sed-4.9/gnulib-tests'
make[4]: Leaving directory '/sources/sed-4.9/gnulib-tests'
make[3]: Leaving directory '/sources/sed-4.9/gnulib-tests'
make[2]: Leaving directory '/sources/sed-4.9/gnulib-tests'
make[1]: Leaving directory '/sources/sed-4.9'
(lfs chroot) root:/sources/sed-4.9# install -d -m755 /usr/share/doc/sed-4.9
(lfs chroot) root:/sources/sed-4.9# install -m644 doc/sed.html /usr/share/doc/sed-4.9

```

-Capitulo 8.32 hecho a las 18:07

-Capitulo 8.33 (gettext-tools 0.26) hecho a las 19:21

```

=====
Testsuite summary for gettext-tools 0.26
=====
# TOTAL: 468
# PASS: 421
# SKIP: 47
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====

```

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-Capitulo 8.34 hecho a las 02:15

```

## ----- ##
## Test results. ##
## ----- ##

712 tests were successful.
64 tests were skipped.
make[3]: Leaving directory '/sources/bison-3.8.2'
make[2]: Leaving directory '/sources/bison-3.8.2'
make[1]: Leaving directory '/sources/bison-3.8.2'
(lfs chroot) root:/sources/bison-3.8.2#

```

-Capitulo 8.35 hecho a las 02:20

```

=====
Testsuite summary for GNU grep 3.12
=====
# TOTAL: 315
# PASS: 276
# SKIP: 39
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====
make[5]: Leaving directory '/sources/grep-3.12/gnulib-tests'
make[4]: Leaving directory '/sources/grep-3.12/gnulib-tests'
make[3]: Leaving directory '/sources/grep-3.12/gnulib-tests'
make[2]: Leaving directory '/sources/grep-3.12/gnulib-tests'
make[1]: Leaving directory '/sources/grep-3.12/gnulib-tests'
make[1]: Entering directory '/sources/grep-3.12'
make[1]: Leaving directory '/sources/grep-3.12'
GEN      public-submodule-commit
(lfs chroot) root:/sources/grep-3.12# _

```

-Capitulo 8.36 hecho a las 02:40

-Capitulo 8.37 hecho a las 06:35

```

=====
Testsuite summary for GNU Libtool 2.5.4
=====
# TOTAL: 6
# PASS: 4
# SKIP: 2
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====

```

-Capitulo 8.38 hecho a las 06:40

```

## ----- ##
## Test results. ##
## ----- ##

All 38 tests were successful.

```

-Capitulo 8.39 hecho a las 06:45

-Capitulo 8.40 hecho a las 07:05

```

=====
Testsuite summary for expat 2.7.3
=====
# TOTAL: 2
# PASS: 2
# SKIP: 0
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====

```

-Capitulo 8.41 hecho a las 07:10

```

=====
Testsuite summary for GNU inetutils 2.6
=====
# TOTAL: 12
# PASS: 11
# SKIP: 1
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====

```

-Capitulo 8.42 hecho a las 07:15

```

$ make install
(Ifs chroot) root:/sources/less-679# make install
./mkinstalldirs /usr/bin /usr/share/man/man1
/usr/bin/install -c less /usr/bin/less
/usr/bin/install -c lesskey /usr/bin/lesskey
/usr/bin/install -c lessecho /usr/bin/lessecho
/usr/bin/install -c -m 644 ./less.nro /usr/share/man/man1/less.1
/usr/bin/install -c -m 644 ./lesskey.nro /usr/share/man/man1/lesskey.1
/usr/bin/install -c -m 644 ./lessecho.nro /usr/share/man/man1/lessecho.1
(Ifs chroot) root:/sources/less-679#

```

-Capitulo 8.43 hecho a las 09:20

```

Test Summary Report
-----
run/todo.t                                (Wstat: 0 Tests: 6 Failed: 0)
  TODO passed: 3-4
Files=2915, Tests=1351054, 2416 wallclock secs (335.39 usr 33.49 sys + 3104.21 cusr 232.34 csys = 3705.43 CPU)
Result: PASS
Finished test run at Thu Dec 25 00:48:41 2025.
(Ifs chroot) root:/sources/perl-5.42.0# _

```

-Capitulo 8.44 hecho a las 09:27

```

(Ifs chroot) root:/sources/XML-Parser-2.47# make test
make[1]: Entering directory '/sources/XML-Parser-2.47/Expat'
"/usr/bin/perl" -MExtUtils::Command::MM -e 'cp_nonempty' -- Expat.bs ../blib/arch/auto/XML/Parser/Expat/Expat.bs 644
Manifesting 1 pod document
make[1]: Leaving directory '/sources/XML-Parser-2.47/Expat'
PERL_DL_NONLAZY=1 "/usr/bin/perl" "-MExtUtils::Command::MM" "-MTest::Harness" "-e" "undef *Test::Harness::Switches; test
*.t
t/astress.t ..... ok
t/cdata.t ..... ok
t/decl.t ..... ok
t/defaulted.t ..... ok
t/encoding.t ..... ok
t/external_ent.t ..... Couldn't load LWP based external entity handler
Switching to file-based external entity handler
(To avoid this message, use NoLWP option to XML::Parser)
t/external_ent.t ..... ok
t/file.t ..... ok
t/file_open_scalar.t .. ok
t/finish.t ..... ok
t/namespaces.t ..... ok
t/parament.t ..... 1/13 Couldn't load LWP based external entity handler
Switching to file-based external entity handler
(To avoid this message, use NoLWP option to XML::Parser)
t/parament.t ..... ok
t/partial.t ..... ok
t/skip.t ..... ok
t/stream.t ..... ok
t/styles.t ..... ok
All tests successful.
Files=15, Tests=140, 3 wallclock secs ( 0.16 usr 0.05 sys + 1.80 cusr 0.34 csys = 2.35 CPU)
Result: PASS
(Ifs chroot) root:/sources/XML-Parser-2.47#

```

-Capitulo 8.45 hecho a las 09:30

```

=====
Testsuite summary for intltool 0.51.0
=====
# TOTAL: 1
# PASS: 1
# SKIP: 0
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====

```

-Capitulo 8.46 hecho a las 10:41

```

## ----- ##
## Test results. ##
## ----- ##

549 tests behaved as expected.
60 tests were skipped.
make[2]: Leaving directory '/sources/autoconf-2.72'
make[1]: Leaving directory '/sources/autoconf-2.72'
(lfs chroot) root:/sources/autoconf-2.72#

```

-Capitulo 8.47 hecho a las 18:57

```

=====
Testsuite summary for GNU Automake 1.18.1
=====
# TOTAL: 2960
# PASS: 2761
# SKIP: 160
# XFAIL: 39
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====

```

-Capitulo 8.48 hecho a las 23:40

```

*****
***
***  OpenSSL has been successfully configured  ***
***
***  If you encounter a problem while building, please open an  ***
***  issue on GitHub <https://github.com/openssl/openssl/issues> ***
***  and include the output from the following command:  ***
***
***      perl configdata.pm --dump  ***
***
***  (If you are new to OpenSSL, you might want to consult the  ***
***  'Troubleshooting' section in the INSTALL.md file first)  ***
***
*****

```

-Capitulo 8.49 hecho a las 23:45

```

FAIL: dwarf_srclang_check
=====
Testsuite summary for elfutils 0.193
=====
# TOTAL: 1
# PASS: 0
# SKIP: 0
# XFAIL: 0
# FAIL: 1
# XPASS: 0
# ERROR: 0
=====
See libdw/test-suite.log for debugging.
Some test(s) failed. Please report this to https://sourceware.org/bugzilla,
together with the test-suite.log file (gzipped) and your system
information. Thanks.
=====

```

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-Capitulo 8.50 hecho a las 05:30

```

-----
Libraries have been installed in:
  /usr/lib/./lib

If you ever happen to want to link against installed libraries
in a given directory, LIBDIR, you must either use libtool, and
specify the full pathname of the library, or use the '-LLIBDIR'
flag during linking and do at least one of the following:
  - add LIBDIR to the 'LD_LIBRARY_PATH' environment variable
    during execution
  - add LIBDIR to the 'LD_RUN_PATH' environment variable
    during linking
  - use the '-Wl,-rpath -Wl,LIBDIR' linker flag
  - have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.
-----

/usr/bin/mkdir -p '/usr/lib/pkgconfig'
/usr/bin/install -c -m 644 libffi.pc '/usr/lib/pkgconfig'
make[3]: Leaving directory '/sources/libffi-3.5.2/x86_64-pc-linux-gnu'
make[2]: Leaving directory '/sources/libffi-3.5.2/x86_64-pc-linux-gnu'
make[1]: Leaving directory '/sources/libffi-3.5.2/x86_64-pc-linux-gnu'
(lfs chroot) root:/sources/libffi-3.5.2# _

```

-Capitulo 8.51 hecho a las 9:15

Obs: en el make test solo fueron succesful 499 de 500 tests.

```

== Tests result: FAILURE then FAILURE ==

10 slowest tests:
- test.test_multiprocessing_spawn.test_processes: 3 min 17 sec
- test.test_multiprocessing_forkserver.test_processes: 2 min 22 sec
- test_regrtest: 2 min 2 sec
- test.test_multiprocessing_spawn.test_misc: 1 min 47 sec
- test.test_concurrent_futures.test_process_pool: 1 min 46 sec
- test_tarfile: 1 min 36 sec
- test_zipimport: 1 min 33 sec
- test.test_multiprocessing_spawn.test_manager: 1 min 32 sec
- test_compile: 1 min 27 sec
- test.test_concurrent_futures.test_deadlock: 1 min 22 sec

28 tests skipped:
  test.test_asyncio.test_windows_events
  test.test_asyncio.test_windows_utils test.test_gdb.test_backtrace
  test.test_gdb.test_cfunction test.test_gdb.test_cfunction_full
  test.test_gdb.test_misc test.test_gdb.test_pretty_print
  test_android test_apple test_dbm_sqlite3 test_devpoll
  test_free_threading test_idle test_kqueue test_launcher
  test_msvcrt test_sqlite3 test_startfile test_tcl test_tkinter
  test_ttk test_ttk_textonly test_turtle test_winapi
  test_winconsoleio test_winreg test_winsound test_wmi

2 tests skipped (resource denied):
  test_peg_generator test_zipfile64

1 re-run test:
  test_strptime

1 test failed:
  test_strptime

449 tests OK.

Total duration: 19 min 26 sec
Total tests: run=44,235 skipped=1,888
Total test files: run=479/480 failed=1 skipped=28 resource_denied=2 rerun=1
Result: FAILURE then FAILURE
make: *** [Makefile:2198: test] Error 2
(lfs chroot) root:/sources/Python-3.13.7#

```

```

The necessary bits to build these optional modules were not found:
_tkinter
To find the necessary bits, look in configure.ac and config.log.

Checked 112 modules (33 built-in, 76 shared, 1 n/a on linux-x86_64, 1 disabled, 1 missing, 0 failed on import)

```

-Capítulo 8.52 hecho a las 9:18

```

(lfs chroot) root:/sources# tar -xzf flit_core-3.12.0.tar.gz
(lfs chroot) root:/sources# cd flit_core-3.12.0
(lfs chroot) root:/sources/flit_core-3.12.0# pip3 wheel -w dist --no-cache-dir --no-build-isolation --no-deps $PWD
Processing /sources/flit_core-3.12.0
  Preparing metadata (pyproject.toml) ... done
Building wheels for collected packages: flit_core
  Building wheel for flit_core (pyproject.toml) ... done
  Created wheel for flit_core: filename=flit_core-3.12.0-py3-none-any.whl size=45594 sha256=15c45ef461fe19ff565a8f484431a5eda34743447f09b3af06469a4280cc72af
  Stored in directory: /tmp/pip-ephem-wheel-cache-ugbZh8xj/wheels/93/2e/39/cbd4768a11a1b386c867978e1745135818c8ef0eda58
Successfully built flit_core
(lfs chroot) root:/sources/flit_core-3.12.0# pip3 install --no-index --find-links dist flit_core
Looking in links: dist
Processing ./dist/flit_core-3.12.0-py3-none-any.whl
Installing collected packages: flit_core
Successfully installed flit_core-3.12.0
(lfs chroot) root:/sources/flit_core-3.12.0#

```

-Capítulo 8.53 hecho a las 9:25



```
(lfs chroot) root:/sources# tar -xzf packaging-25.0.tar.gz
(lfs chroot) root:/sources# cd packaging-25.0
(lfs chroot) root:/sources/packaging-25.0# pip3 wheel -w dist --no-cache-dir --no-build-isolation --no-deps $PWD
Processing /sources/packaging-25.0
  Preparing metadata (pyproject.toml) ... done
Building wheels for collected packages: packaging
  Building wheel for packaging (pyproject.toml) ... done
  Created wheel for packaging: filename=packaging-25.0-py3-none-any.whl size=66469 sha256=5a5aa445e68cdc812e844831be98c89a16443b7a68cb7cabab9af7613a4d32f2
  Stored in directory: /tmp/pip-ephem-wheel-cache-8jop6444/wheels/75/65/3a/8aebad6992b02a07e4fa7b0d068ae833e505e0d766e6b4812
Successfully built packaging
(lfs chroot) root:/sources/packaging-25.0# pip3 install --no-index --find-links dist packaging
Looking in links: dist
Processing ./dist/packaging-25.0-py3-none-any.whl
Installing collected packages: packaging
Successfully installed packaging-25.0
(lfs chroot) root:/sources/packaging-25.0#
```

-Capitulo 8.54 hecho a las 9:30

-Capitulo 8.55 hecho a las 9:35

```
(lfs chroot) root:/sources# tar -xzf setuptools-00.9.0.tar.gz
(lfs chroot) root:/sources# cd setuptools-00.9.0
(lfs chroot) root:/sources/setuptools-00.9.0# pip3 wheel -w dist --no-cache-dir --no-build-isolation --no-deps $PWD
Processing /sources/setuptools-00.9.0
  Preparing metadata (pyproject.toml) ... done
Building wheels for collected packages: setuptools
  Building wheel for setuptools (pyproject.toml) ... done
  Created wheel for setuptools: filename=setuptools-00.9.0-py3-none-any.whl size=1201486 sha256=2eaab171e970e184a1814340e80bf18e1e6c72d3409c34c0efbe79f35470b662
  Stored in directory: /tmp/pip-ephem-wheel-cache-n9yrlkjg/wheels/b2/07/c0/2b275b554f5ad8af65f99b37d7a679b1155088a0e18d6fbed4
Successfully built setuptools
(lfs chroot) root:/sources/setuptools-00.9.0# pip3 install --no-index --find-links dist setuptools
Looking in links: dist
Processing ./dist/setuptools-00.9.0-py3-none-any.whl
Installing collected packages: setuptools
Successfully installed setuptools-00.9.0
(lfs chroot) root:/sources/setuptools-00.9.0#
```

-Capitulo 8.56 hecho a las 10:00

```
(lfs chroot) root:/sources/ninja-1.13.1# install -vm755 ninja /usr/bin/
'ninja' -> '/usr/bin/ninja'
(lfs chroot) root:/sources/ninja-1.13.1# install -vDm644 misc/bash-completion /usr/share/bash-completion/completions/ninja
'misc/bash-completion' -> '/usr/share/bash-completion/completions/ninja'
(lfs chroot) root:/sources/ninja-1.13.1# install -vDm644 misc/zsh-completion /usr/share/zsh/site-functions/_ninja
install: creating directory '/usr/share/zsh'
install: creating directory '/usr/share/zsh/site-functions'
'misc/zsh-completion' -> '/usr/share/zsh/site-functions/_ninja'
(lfs chroot) root:/sources/ninja-1.13.1# cd ..
(lfs chroot) root:/sources# rm -Rf ninja-1.13.1
```

-Capitulo 8.57 hecho a las 10:03

```
(lfs chroot) root:/sources# tar -xzf meson-1.8.3.tar.gz
(lfs chroot) root:/sources# cd meson-1.8.3
(lfs chroot) root:/sources/meson-1.8.3#
(lfs chroot) root:/sources/meson-1.8.3# pip3 wheel -w dist --no-cache-dir --no-build-isolation --no-deps $PWD
Processing /sources/meson-1.8.3
  Preparing metadata (pyproject.toml) ... done
Building wheels for collected packages: meson
  Building wheel for meson (pyproject.toml) ... done
  Created wheel for meson: filename=meson-1.8.3-py3-none-any.whl size=1014616 sha256=b9131f3924f6f2d5949b6392a37665e88228fab63919d0631337783bc3a6b8d0
  Stored in directory: /tmp/pip-ephem-wheel-cache-ux6pf4e4/wheels/62/54/c4/1f8a0591dc9b351b9583369e764c1a9417f6cf6b549106225e5
Successfully built meson
(lfs chroot) root:/sources/meson-1.8.3# pip3 install --no-index --find-links dist meson
Looking in links: dist
Processing ./dist/meson-1.8.3-py3-none-any.whl
Installing collected packages: meson
Successfully installed meson-1.8.3
(lfs chroot) root:/sources/meson-1.8.3# install -vDm644 data/shell-completions/bash/meson /usr/share/bash-completion/completions/meson
'data/shell-completions/bash/meson' -> '/usr/share/bash-completion/completions/meson'
(lfs chroot) root:/sources/meson-1.8.3# install -vDm644 data/shell-completions/zsh/_meson /usr/share/zsh/site-functions/_meson
'data/shell-completions/zsh/_meson' -> '/usr/share/zsh/site-functions/_meson'
(lfs chroot) root:/sources/meson-1.8.3# cd ..
(lfs chroot) root:/sources# rm -Rf meson-1.8.3
```

-Capitulo 8.58 hecho a las 10:15

```

(lfs chroot) root:/sources/kmod-34.2/build# ninja
[36/36] Generating rmmmod with a custom command
(lfs chroot) root:/sources/kmod-34.2/build# ninja install
[0/1] Installing files
Installing libkmod.so.2.5.1 to /usr/lib
Installing kmod to /usr/bin
Installing /sources/kmod-34.2/libkmod/libkmod.h to /usr/include
Installing new directory /etc/depmod.d
Installing new directory /etc/modprobe.d
Installing new directory /usr/lib/depmod.d
Installing new directory /usr/lib/modprobe.d
Installing /sources/kmod-34.2/shell-completion/bash/insmod to /usr/share/bash-completion/completions
Installing /sources/kmod-34.2/shell-completion/bash/lsmmod to /usr/share/bash-completion/completions
Installing /sources/kmod-34.2/shell-completion/bash/rmmmod to /usr/share/bash-completion/completions
Installing /sources/kmod-34.2/shell-completion/fish/insmod.fish to /usr/share/fish/vendor_functions.d
Installing /sources/kmod-34.2/shell-completion/fish/lsmmod.fish to /usr/share/fish/vendor_functions.d
Installing /sources/kmod-34.2/shell-completion/fish/rmmmod.fish to /usr/share/fish/vendor_functions.d
Installing /sources/kmod-34.2/shell-completion/zsh/_insmod to /usr/share/zsh/site-functions
Installing /sources/kmod-34.2/shell-completion/zsh/_lsmmod to /usr/share/zsh/site-functions
Installing /sources/kmod-34.2/shell-completion/zsh/_rmmmod to /usr/share/zsh/site-functions
Installing /sources/kmod-34.2/shell-completion/bash/kmod to /usr/share/bash-completion/completions
Installing /sources/kmod-34.2/build/meson-private/libkmod.pc to /usr/lib/pkgconfig
Installing /sources/kmod-34.2/build/meson-private/kmod.pc to /usr/share/pkgconfig
Installing symlink pointing to libkmod.so.2.5.1 to /usr/lib/libkmod.so.2
Installing symlink pointing to libkmod.so.2 to /usr/lib/libkmod.so
Running custom install script '/usr/bin/sh -c mkdir -p "$DESTDIR/usr/sbin"'
Running custom install script '/usr/bin/sh -c ln -sfr "$DESTDIR/usr/bin/kmod" "$DESTDIR/usr/sbin/depmod"'
Running custom install script '/usr/bin/sh -c ln -sfr "$DESTDIR/usr/bin/kmod" "$DESTDIR/usr/sbin/insmod"'
Running custom install script '/usr/bin/sh -c ln -sfr "$DESTDIR/usr/bin/kmod" "$DESTDIR/usr/sbin/lsmmod"'
Running custom install script '/usr/bin/sh -c ln -sfr "$DESTDIR/usr/bin/kmod" "$DESTDIR/usr/sbin/modinfo"'
Running custom install script '/usr/bin/sh -c ln -sfr "$DESTDIR/usr/bin/kmod" "$DESTDIR/usr/sbin/modprobe"'
Running custom install script '/usr/bin/sh -c ln -sfr "$DESTDIR/usr/bin/kmod" "$DESTDIR/usr/sbin/rmmmod"'
(lfs chroot) root:/sources/kmod-34.2/build#

```

-Capitulo 8.59 hecho a las 11:45

```

=====
Testsuite summary for GNU coreutils 9.7
=====
# TOTAL: 36
# PASS: 19
# SKIP: 15
# XFAIL: 0
# FAIL: 2
# XPASS: 0
# ERROR: 0
=====
See ./tests/test-suite.log for debugging.
Some test(s) failed. Please report this to bug-coreutils@gnu.org,
together with the test-suite.log file (gzipped) and your system
information. Thanks.
=====

```

```

FAIL: tests/chroot/chroot-credentials.sh
selinux.sh: skipped test: this build lacks SELinux support
SKIP: tests/misc/selinux.sh
PASS: tests/truncate/truncate-owned-by-other.sh
writable-under-readonly.sh: skipped test: temporarily disabled
SKIP: tests/mkdir/writable-under-readonly.sh
smack-root.sh: skipped test: this system lacks SMACK support
SKIP: tests/mkdir/smack-root.sh
hardlink-case.sh: skipped test: failed to create hfs file system
SKIP: tests/mv/hardlink-case.sh
FAIL: tests/mv/sticky-to-xpart.sh

```

```

=====
Testsuite summary for GNU coreutils 9.7
=====
# TOTAL: 527
# PASS: 472
# SKIP: 55
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====

```

-Capitulo 8.60 hecho a las 11:50

```

=====
Testsuite summary for GNU diffutils 3.12
=====
# TOTAL: 344
# PASS: 299
# SKIP: 45
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====

```

-Capitulo 8.61 hecho a las 12:05

```

===== Done with machine-specific tests =====
make[3]: Entering directory '/sources/gawk-5.3.2/test'
ALL TESTS PASSED
make[3]: Leaving directory '/sources/gawk-5.3.2/test'
make[2]: Leaving directory '/sources/gawk-5.3.2/test'
make[1]: Leaving directory '/sources/gawk-5.3.2'
(lfs chroot) root:/sources/gawk-5.3.2#

```

-Capitulo 8.62 hecho a las 12:25

```

=====
Testsuite summary for GNU findutils 4.10.0
=====
# TOTAL: 23
# PASS: 21
# SKIP: 2
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====

```

-Capitulo 8.63 hecho a las 12:40

```

=====
Testsuite summary for GNU roff 1.23.0
=====
# TOTAL: 164
# PASS: 151
# SKIP: 11
# XFAIL: 2
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====
make[3]: Leaving directory '/sources/groff-1.23.0'
make[2]: Leaving directory '/sources/groff-1.23.0'
make[1]: Leaving directory '/sources/groff-1.23.0'
(lfs chroot) root:/sources/groff-1.23.0#

```

-Capítulo 8.64 hecho a las 19:30

```

=====
Testsuite summary for gzip 1.14
=====
# TOTAL: 30
# PASS: 29
# SKIP: 1
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====
make[4]: Leaving directory '/sources/gzip-1.14/tests'
make[3]: Leaving directory '/sources/gzip-1.14/tests'
make[2]: Leaving directory '/sources/gzip-1.14/tests'
make[1]: Leaving directory '/sources/gzip-1.14'
(lfs chroot) root:/sources/gzip-1.14#

```

-Capítulo 8.65 hecho a las 20:35

-Capítulo 8.66 hecho a las 20:40

-Capítulo 8.67 hecho a las 20:44

-Capítulo 8.68 hecho a las 20:47

-Capítulo 8.69 hecho a las 21:00

```

1431 Tests in 133 Categories Complete ... No Failures :-)

=====
Regression PASSED: GNU Make 4.4.1 (x86_64-pc-linux-gnu) built with gcc
=====
make[2]: Leaving directory '/sources/make-4.4.1'
make[1]: Leaving directory '/sources/make-4.4.1'
(lfs chroot) root:/sources/make-4.4.1#

```

-Capítulo 8.70 hecho a las 21:10

```

=====
Testsuite summary for GNU patch 2.8
=====
# TOTAL: 49
# PASS: 45
# SKIP: 2
# XFAIL: 2
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====
make[4]: Leaving directory '/sources/patch-2.8/tests'
make[3]: Leaving directory '/sources/patch-2.8/tests'
make[2]: Leaving directory '/sources/patch-2.8/tests'
make[2]: Entering directory '/sources/patch-2.8'
make[2]: Leaving directory '/sources/patch-2.8'
make[1]: Leaving directory '/sources/patch-2.8'
(lfs chroot) root:/sources/patch-2.8#

```

-Capitulo 8.71 hecho a las 21:30

```

233: capabilities: binary store/restore          FAILED (capabs_raw01.at:28)

One top level

234: tar --one-top-level                        ok
235: tar --one-top-level --show-transformed      ok
236: tar --one-top-level --transform            ok
237: tar --one-top-level --transform            ok
238: tar --one-top-level restoring permissions   ok

Star tests

239: gtarfail                                  skipped (gtarfail.at:26)
240: gtarfail2                                skipped (gtarfail2.at:26)
241: multi-fail                                skipped (multi-fail.at:26)
242: ustar-big-2g                              skipped (ustar-big-2g.at:26)
243: ustar-big-8g                              skipped (ustar-big-8g.at:26)
244: pax-big-10g                               skipped (pax-big-10g.at:26)

## ----- ##
## Test results. ##
## ----- ##

ERROR: 224 tests were run,
1 failed unexpectedly.
20 tests were skipped.
## ----- ##
## testsuite.log was created. ##
## ----- ##

Please send `tests/testsuite.log' and all information you think might help:

    To: <bug-tar@gnu.org>
    Subject: [GNU tar 1.35] testsuite: 233 failed

You may investigate any problem if you feel able to do so, in which
case the test suite provides a good starting point. Its output may
be found below `tests/testsuite.dir'.

```

-Capitulo 8.72 hecho a las 21:50

-Capitulo 8.73 hecho a las 22:30

```

(lfs chroot) root:/sources/vim-9.1.1629# chown -R tester .
(lfs chroot) root:/sources/vim-9.1.1629# sed '/test_plugin_globs/d' -i src/testdir/Make_all.mak
(lfs chroot) root:/sources/vim-9.1.1629# su tester -c "TERM=xterm-256color LANG=en_US.UTF-8 make -j1 test" \
> &> vim-test.log
(lfs chroot) root:/sources/vim-9.1.1629# make install

```

```
(lfs chroot) root:/sources/vim-9.1.1629# ln -sv vim /usr/bin/vi
/usr/bin/vi' -> 'vim'
(lfs chroot) root:/sources/vim-9.1.1629# for L in /usr/share/man/{,*/}man1/vim.1; do
> ln -sv vim.1 $(dirname $L)/vi.1
> done
/usr/share/man/man1/vi.1' -> 'vim.1'
/usr/share/man/da.ISO8859-1/man1/vi.1' -> 'vim.1'
/usr/share/man/da.UTF-8/man1/vi.1' -> 'vim.1'
/usr/share/man/da/man1/vi.1' -> 'vim.1'
/usr/share/man/de.ISO8859-1/man1/vi.1' -> 'vim.1'
/usr/share/man/de.UTF-8/man1/vi.1' -> 'vim.1'
/usr/share/man/de/man1/vi.1' -> 'vim.1'
/usr/share/man/fr.ISO8859-1/man1/vi.1' -> 'vim.1'
/usr/share/man/fr.UTF-8/man1/vi.1' -> 'vim.1'
/usr/share/man/fr/man1/vi.1' -> 'vim.1'
/usr/share/man/it.ISO8859-1/man1/vi.1' -> 'vim.1'
/usr/share/man/it.UTF-8/man1/vi.1' -> 'vim.1'
/usr/share/man/it/man1/vi.1' -> 'vim.1'
/usr/share/man/ja/man1/vi.1' -> 'vim.1'
/usr/share/man/pl.ISO8859-2/man1/vi.1' -> 'vim.1'
/usr/share/man/pl.UTF-8/man1/vi.1' -> 'vim.1'
/usr/share/man/pl/man1/vi.1' -> 'vim.1'
/usr/share/man/ru.KOI8-R/man1/vi.1' -> 'vim.1'
/usr/share/man/ru.UTF-8/man1/vi.1' -> 'vim.1'
/usr/share/man/tr.ISO8859-9/man1/vi.1' -> 'vim.1'
/usr/share/man/tr.UTF-8/man1/vi.1' -> 'vim.1'
/usr/share/man/tr/man1/vi.1' -> 'vim.1'
(lfs chroot) root:/sources/vim-9.1.1629# ln -sv ../vim/vim91/doc /usr/share/doc/vim-9.1.1629
/usr/share/doc/vim-9.1.1629' -> '../vim/vim91/doc'
(lfs chroot) root:/sources/vim-9.1.1629#
(lfs chroot) root:/sources/vim-9.1.1629# cat > /etc/vimrc << "EOF"
> " Begin /etc/vimrc
> " Ensure defaults are set before customizing settings, not after
> source $VIMRUNTIME/defaults.vim
> let skip_defaults_vim=1
> set nocompatible
> set backspace=2
> set mouse=
> syntax on
> if (&term == "xterm") || (&term == "putty")
> set background=dark
> endif
> " End /etc/vimrc
> EOF
(lfs chroot) root:/sources/vim-9.1.1629# vim -c ':options'
```

-Capítulo 8.74 hecho a las 22:57

```
(lfs chroot) root:/sources# cd markupsafe-3.0.2
(lfs chroot) root:/sources/markupsafe-3.0.2# pip3 wheel -w dist --no-cache-dir --no-build-isolation --no-deps $PWD
Processing /sources/markupsafe-3.0.2
  Preparing metadata (pyproject.toml) ... done
Building wheels for collected packages: MarkupSafe
  Building wheel for MarkupSafe (pyproject.toml) ... done
  Created wheel for MarkupSafe: filename=markupsafe-3.0.2-cp313-cp313-linux_x86_64.whl size=22948 sha256=28a395cbfe36bc425894a8764fa4bd643e12262a2adb7d6c38449c39d39ce9d5
  Stored in directory: /tmp/pip-ephem-wheel-cache-s0t133h_/wheels/fb/ba/92/578e9b60b3e2eb767cf4035534317a66d3aa9eda6df8c84ad
Successfully built MarkupSafe
(lfs chroot) root:/sources/markupsafe-3.0.2# pip3 install --no-index --find-links dist MarkupSafe
Looking in links: dist
Processing ./dist/markupsafe-3.0.2-cp313-cp313-linux_x86_64.whl
Installing collected packages: MarkupSafe
Successfully installed MarkupSafe-3.0.2
(lfs chroot) root:/sources/markupsafe-3.0.2# cd ..
(lfs chroot) root:/sources# rm -rf markupsafe-3.0.2
```

-Capítulo 8.75 hecho a las 23:30

```
(lfs chroot) root:/sources# tar -xzf jinja2-3.1.6.tar.gz
(lfs chroot) root:/sources# cd jinja2-3.1.6
(lfs chroot) root:/sources/jinja2-3.1.6# pip3 wheel -w dist --no-cache-dir --no-build-isolation --no-deps $PWD
Processing /sources/jinja2-3.1.6
  Preparing metadata (pyproject.toml) ... done
Building wheels for collected packages: Jinja2
  Building wheel for Jinja2 (pyproject.toml) ... done
  Created wheel for Jinja2: filename=jinja2-3.1.6-py3-none-any.whl size=134897 sha256=e5582add22854df03c370ef549e88a22cef3fcf452788987c5f1bc72b5c3c9ba
  Stored in directory: /tmp/pip-ephem-wheel-cache-5m5ly546/wheels/4f/7a/c5/48cc5c37ed16befec3c22c6c9c98d23bb7881ce4144fa138885
Successfully built Jinja2
(lfs chroot) root:/sources/jinja2-3.1.6# pip3 install --no-index --find-links dist Jinja2
Looking in links: dist
Processing ./dist/jinja2-3.1.6-py3-none-any.whl
Requirement already satisfied: MarkupSafe>=2.0 in /usr/lib/python3.13/site-packages (from Jinja2) (3.0.2)
Installing collected packages: Jinja2
Successfully installed Jinja2-3.1.6
(lfs chroot) root:/sources/jinja2-3.1.6# cd ..
(lfs chroot) root:/sources# rm -rf jinja2-3.1.6
```

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-Capitulo 8.76 hecho a las 01:00

```
Found ninja-1.13.1 at /usr/bin/ninja
(lfs chroot) root:/sources/systemd-257.8/build# ninja
[2289/2289] Generating export-dbus-interfaces with a custom command
(lfs chroot) root:/sources/systemd-257.8/build# echo 'NAME="Linux From Scratch"' > /etc/os-release
(lfs chroot) root:/sources/systemd-257.8/build#
```

```

Features
enabled
: ACL, IMA, IPE, SMACK, blkid, libfdisk, openssl, zstd, lz4, xz, zlib, bzip2, backlight, binfmt, coredump, efi, emu
ironment.d, hibernate, hostnamed, hwdb, initrd, kernel-install, locale, logind, machined, mountfsd, networkd, nssresourced, nss-myhostname, nss-my machines, nss-
resolve, nss-systemd, oomd, portabled, pstore, quotacheck, randomseed, report, resolve, rftkill, susext, systemd-analyze, storagectl, timedated, timesyncd, tmpfil
es, uconsole, xdg-autostart, dmi, idn, polkit, kmod, tpm, SysV compat, utmp, adm group, wheel group, gshadow, fuzz tests, link-udev-shared, link-systemctl-share
d, link-networkd-shared, link-timesyncd-shared, link-journaltctl-shared, link-boot-shared, link-portabled-shared, cryptolib(openssl), DNS-over-TLS(openssl)
disabled
: AUDIT, AppArmor, PAM, SECCOMP, SELinux, elfutils, gcrypt, gnutils, libarchive, libbpf, libcryptsetup, libcryptsetu
p-plugins, libcurl, libfdo2, libidn, libidn2, libipsec, microhttpd, p11kit, passwdie, pcre2, pqauality, qrencode, tpm2, x86common, bootloader, bpf-framework, f
irmware, kernel, importd, sysupdate, sysupdate, users, userdb, vmspan, legacy-pki, xenctrl, dbus, glib, man pages, html pages, man page indices, compat-nu
tile-uid-boundaries, lsdconfig, debug hashmap, debug mmap cache, debug siphash, trace logging, slow tests, install tests, first-boot-full-preset, fexxxv, standa
lone-binaries, coverage, static-libsystemd, static-libudev

User defined options
buildtype
: release
default-dnssec
: no
dev-icev-mode
: 0660
docdir
: /usr/share/doc/systemd-257.8
firstboot
: false
homed
: disabled
install-tests
: false
lsdconfig
: false
man
: disabled
mode
: release
mohady-group
: nogroup
pamconfdir
: no
prefix
: /usr
rpmmacrodir
: no
sysupdate
: disabled
sysusers
: false
ukify
: disabled
userdb
: false

Found ninja-1.13.1 at /usr/bin/ninja
(lfs.chroot) root:/sources/systemd-257.8/build#

```

```
Found ninja-1.13.1 at /usr/bin/ninja
(lfs chroot) root:/sources/systemd-257.8/build#
(lfs chroot) root:/sources/systemd-257.8/build#
(lfs chroot) root:/sources/systemd-257.8/build# ninja
[2289/2289] Generating export-dbus-interfaces with a custom command
(lfs chroot) root:/sources/systemd-257.8/build#
```

-Capitulo 8.77 hecho a las 01:00

```
[172/172] Linking target test/test-misc-internals
(lfs chroot) root:/sources/dbus-1.16.2/build# ninja test
[0/1] Running all tests
1/6 atomic                OK                0.02s    2 subtests passed
2/6 misc-internals        OK                0.92s    7 subtests passed
3/6 shell                 OK                0.03s   11 subtests passed
4/6 strings               OK                0.04s    1 subtests passed
5/6 printf                OK                0.06s    8 subtests passed
6/6 hash                  OK                4.29s    1 subtests passed

Ok:                        6
Fail:                      0

Full log written to /sources/dbus-1.16.2/build/meson-logs/testlog.txt
```

-Capitulo 8.78 hecho a las 01:00

```

=====
Testsuite summary for man-db 2.13.1
=====
# TOTAL: 12
# PASS: 12
# SKIP: 0
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====

```

-Capítulo 8.79 hecho a las 01:00

```

=====
Testsuite summary for procs-ng 4.0.5
=====
# TOTAL: 8
# PASS: 8
# SKIP: 0
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====

```

-Capítulo 8.80 hecho a las 01:00

```

-----
1 tests of 336 FAILED

kill/decode
-----

```

-Capítulo 8.81 hecho a las 01:00

```

390 tests succeeded      1 tests failed
Tests failed: m_assume_storage_prezeroed
make[1]: *** [Makefile:403: test_post] Error 1
make[1]: Leaving directory '/sources/e2fsprogs-1.47.3/build/tests'
make: *** [Makefile:430: check-recursive] Error 1
(lfs chroot) root:/sources/e2fsprogs-1.47.3/build# _

```

## 8.81.2. Configuring E2fsprogs

`/etc/mke2fs.conf` contains the default value of various command line options of **mke2fs**. You may edit the file to make the default values suitable for your needs. For example, some utilities (not in LFS or BLFS) cannot recognize a `ext4` file system with `metadata_csum_seed` feature enabled. If you need such a utility, you may remove the feature from the default `ext4` feature list with the command:

```
sed 's/metadata_csum_seed, //' -i /etc/mke2fs.conf
```

Read the man page `mke2fs.conf(5)` for details.



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## Capítulo 9

-Capítulo 9.2 hecho a las 10:30.

- Configuración General de Redes.
- Comandos principales ejecutados: ln, cat, echo.

-Capítulo 9.3 es de lectura, no se realizó nada.

-Capítulo 9.4 no realizado porque no se estará manejando dispositivos.

-Capítulo 9.5 hecho días más tarde debido a que el comando timedatectl no funciona dentro del entorno chroot.

- Configuración del System Clock.
- Comandos principales ejecutados: cat, timedatectl, systemctl disable.

-Capítulo 9.6 hecho días más tarde debido a que el comando localectl no funciona dentro del entorno chroot.

- Configuración de la Linux Console.
- Comandos principales ejecutados: echo, cat, localectl.

-Capítulo 9.7 hecho días más tarde debido a que el comando localectl no funciona dentro del entorno chroot.

- Configuración del System Locale.
- Comandos principales ejecutados: locale, cat > /etc/locale.conf << "EOF", cat > /etc/profile << "EOF", localectl.

-Capítulo 9.8 hecho a las 10:40.

- Creación del archivo /etc/inputrc.
- Comandos principales ejecutados: cat > /etc/inputrc << "EOF".
- Obs: implementado el script proporcionado por el manual.

-Capítulo 9.9 hecho a las 10:50.

- Creación del archivo /etc/shells.
- Comandos principales ejecutados: cat > /etc/shells << "EOF".
- Obs: implementado el script proporcionado por el manual. Modificar en el futuro para la implementación de la Shell.

```

(lfs chroot) root:/sources# cat > /etc/inputrc << "EOF"
> # Begin /etc/inputrc
> # Modified by Chris Lynn <roryo@roryo.dynup.net>
> # Allow the command prompt to wrap to the next line
> set horizontal-scroll-mode Off
> # Enable 8-bit input
> set meta-flag On
> set input-meta On
> # Turns off 8th bit stripping
> set convert-meta Off
> # Keep the 8th bit for display
> set output-meta On
> # none, visible or audible
> set bell-style none
> # All of the following map the escape sequence of the value
> # contained in the 1st argument to the readline specific functions
> "\eOd": backward-word
> "\eOc": forward-word
> # for linux console
> "\eI1~": beginning-of-line
> "\eI4~": end-of-line
> "\eI5~": beginning-of-history
> "\eI6~": end-of-history
> "\eI3~": delete-char
> "\eI2~": quoted-insert
> # for xterm
> "\eOH": beginning-of-line
> "\eOF": end-of-line
> # for Konsole
> "\eIH": beginning-of-line
> "\eIF": end-of-line
> # End /etc/inputrc
> EOF
(lfs chroot) root:/sources# cat > /etc/shells << "EOF"
> # Begin /etc/shells
> /bin/sh
> /bin/bash
> # End /etc/shells
> EOF
(lfs chroot) root:/sources#

```

-Capítulo 9.10 hecho a las 11:30.

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## Capítulo 10

-Capítulo 10.2 hecho a las 00:58

- Creación del archivo /etc/fstab.
- Comandos principales ejecutados: cat > /etc/fstab << "EOF".

```

(lfs chroot) root:/sources# cat > /etc/fstab << "EOF"
> # Begin /etc/fstab
> # file system mount-point type options dump fsck
> # order
> PARTUUID=88f184ce-b5d4-4c99-8d7c-0a4ebe2ce701 / ext4 defaults 1 1
> # End /etc/fstab
> EOF
(lfs chroot) root:/sources# tar -xf linux-6.16.1.tar.xz
(lfs chroot) root:/sources# cd linux-6.16.1
(lfs chroot) root:/sources/linux-6.16.1# make mrproper
(lfs chroot) root:/sources/linux-6.16.1# make defconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/kconfig/conf.o
HOSTCC scripts/kconfig/confdata.o
HOSTCC scripts/kconfig/expr.o
LEX scripts/kconfig/lexer.lex.c
YACC scripts/kconfig/parser.tab.[ch]
HOSTCC scripts/kconfig/lexer.lex.o
HOSTCC scripts/kconfig/menu.o
HOSTCC scripts/kconfig/parser.tab.o
HOSTCC scripts/kconfig/preprocess.o
HOSTCC scripts/kconfig/symbol.o
HOSTCC scripts/kconfig/util.o
HOSTLD scripts/kconfig/conf
*** Default configuration is based on 'x86_64_defconfig'
#
# configuration written to .config
#
(lfs chroot) root:/sources/linux-6.16.1# make menuconfig
HOSTCC scripts/kconfig/mconf.o
HOSTCC scripts/kconfig/lxdialog/checklist.o
HOSTCC scripts/kconfig/lxdialog/inputbox.o
HOSTCC scripts/kconfig/lxdialog/menubox.o
HOSTCC scripts/kconfig/lxdialog/textbox.o
HOSTCC scripts/kconfig/lxdialog/util.o
HOSTCC scripts/kconfig/lxdialog/yesno.o
HOSTCC scripts/kconfig/mnconf-common.o
HOSTLD scripts/kconfig/mconf
configuration written to .config

*** End of the configuration.
*** Execute 'make' to start the build or try 'make help'.

(lfs chroot) root:/sources/linux-6.16.1#

```

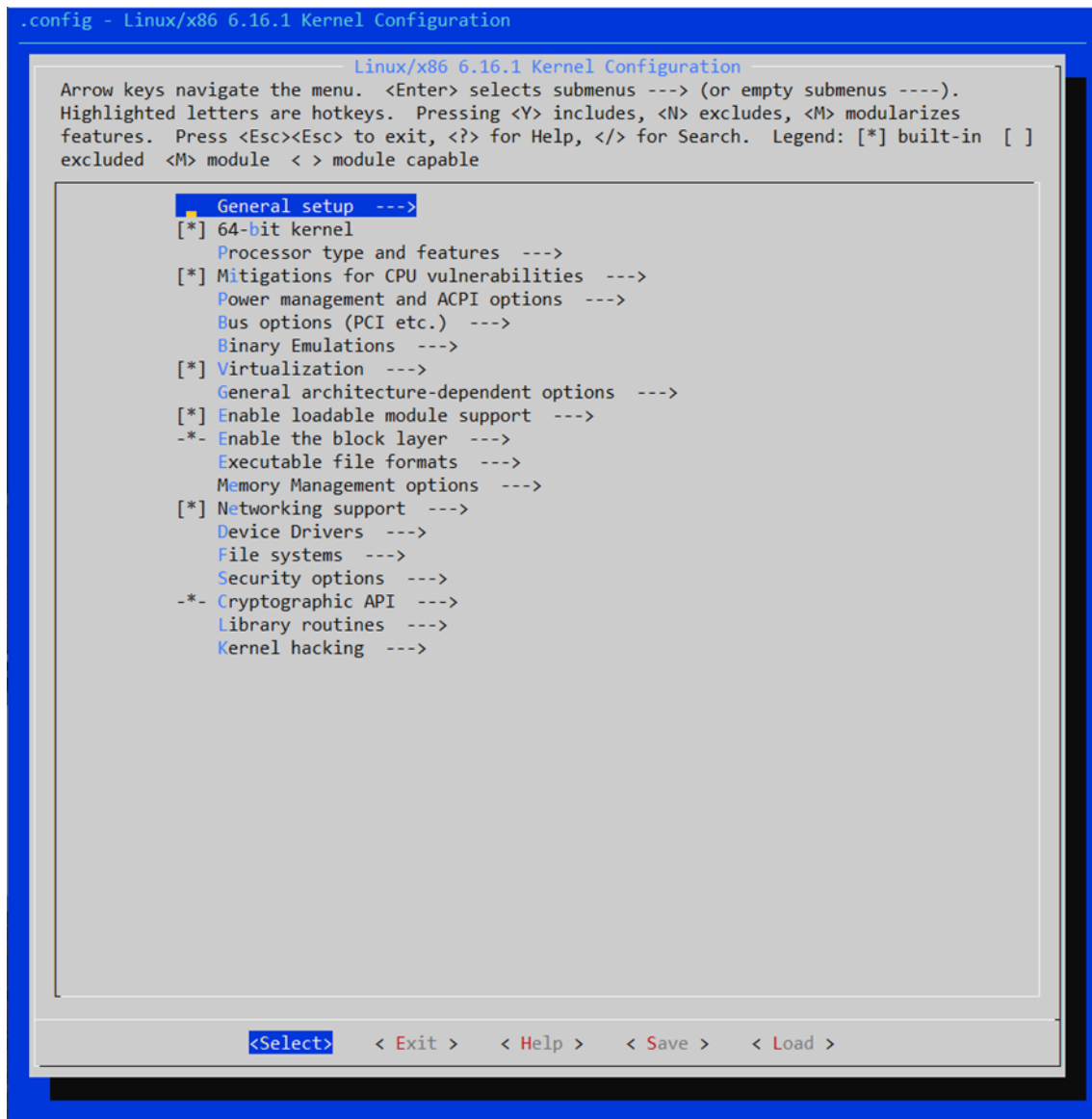
```

# Begin /etc/fstab
# file system mount-point type options dump fsck
# order
PARTUUID=88f184ce-b5d4-4c99-8d7c-0a4ebe2ce701 / ext4 defaults 1 1
UUID=42f11214-c2d3-4843-b54f-67e6d48febba none swap sw 0 0
# End /etc/fstab
~
~

```

-Capítulo 10.3 hecho a las 16:00

- Instalación del Linux Kernel.
- Comandos principales ejecutados: tar, make mrproper, make menuconfig, make defconfig, make, cp, install, cat.



-Capítulo 10.4 hecho a las 21:00.

- Uso de GRUB para configurar el proceso de arranque.
- Comandos principales ejecutados: cp, grub-mkrescue, xorriso, grub-install /dev/sdb, cat > /boot/grub/grub.cfg << "EOF".

```
# Begin /boot/grub/grub.cfg
set default=0
set timeout=5

insmod part_gpt
insmod ext2
set gfxpayload=1024x768x32

search --fs-uuid --set=root afcecafd-4083-4d89-892f-4dc5cb1b0896

menuentry "GNU/Linux, Linux 6.16.1-lfs-12.4-systemd" {
    linux /boot/vmlinuz-6.16.1-lfs-12.4-systemd root=PARTUUID=88f184ce-b5d4-4c99-8d7c-0a4ebe2ce701 ro
}
```

## Capítulo 11

-Hecho a las 04:00 del 31/12/2025.

- Reboot del sistema.
- Comandos principales ejecutados: echo, cat, logout, umount, mountpoint, reboot.

## Configuración de un Servicio en Red

MAC Address: 08:00:27:7b:0b:22

Configuración de systemd-network (DHCP)

### Instalación de OpenSSH

a) Descargamos el paquete

```
wget --no-check-certificate  
https://ftp.openbsd.org/pub/OpenBSD/OpenSSH/portable/openssh-10.0p1.tar.gz
```

b) Desempaquetamos el archivo

```
tar -xvf openssh-10.0p1.tar.gz
```

c) Ejecutamos el siguiente comando

```
install -v -g sys -m700 -d /var/lib/ssh &&  
groupadd -g 50 ssh &&  
useradd -c 'ssh PrivSep' \  
-d /var/lib/ssh \  
-g ssh \  
-s /bin/false \  
-u 50 ssh
```

d) Instalamos OpenSSH ejecutando los siguientes comandos

```
./configure --prefix=/usr \  
--sysconfdir=/etc/ssh \  
--with-privsep-path=/var/lib/ssh \  
--with-default-path=/usr/bin \  
--with-superuser-path=/usr/sbin:/usr/bin \  
--with-pid-dir=/run &&  
make
```

e) Ejecutamos el siguiente comando para probar los resultados

```
make -j1 tests
```

f) Como usuario root ejecutamos

```

/usr/bin/install -c -m 644 sftp-server.8.out /usr/share/man/man8/sftp-server.8
/usr/bin/install -c -m 644 ssh-keysign.8.out /usr/share/man/man8/ssh-keysign.8
/usr/bin/install -c -m 644 ssh-pkcs11-helper.8.out /usr/share/man/man8/ssh-pkcs11-helper.8
/usr/bin/install -c -m 644 ssh-sk-helper.8.out /usr/share/man/man8/ssh-sk-helper.8
/usr/bin/mkdir -p /etc/ssh
/etc/ssh/ssh_config already exists, install will not overwrite
/etc/ssh/sshd_config already exists, install will not overwrite
/etc/ssh/moduli already exists, install will not overwrite
/usr/sbin/sshd -t -f /etc/ssh/sshd_config
removed '/usr/bin/ssh-copy-id'
'contrib/ssh-copy-id' -> '/usr/bin/ssh-copy-id'
'contrib/ssh-copy-id.1' -> '/usr/share/man/man1/ssh-copy-id.1'
install: creating directory '/usr/share/doc/openssh-10.0p1'
'INSTALL' -> '/usr/share/doc/openssh-10.0p1/INSTALL'
'LICENCE' -> '/usr/share/doc/openssh-10.0p1/LICENCE'
'OVERVIEW' -> '/usr/share/doc/openssh-10.0p1/OVERVIEW'
'README' -> '/usr/share/doc/openssh-10.0p1/README'
'README.dns' -> '/usr/share/doc/openssh-10.0p1/README.dns'
'README.md' -> '/usr/share/doc/openssh-10.0p1/README.md'
'README.platform' -> '/usr/share/doc/openssh-10.0p1/README.platform'
'README.privsep' -> '/usr/share/doc/openssh-10.0p1/README.privsep'
'README.tun' -> '/usr/share/doc/openssh-10.0p1/README.tun'
-bash-5.3# _

```

g) Escribimos el archivo /etc/systemd/system/sshd.service (sugerido por la IA)

h) Ejecutamos los siguientes comandos

```
systemctl daemon-reload
```

```
systemctl enable --now sshd
```

```
systemctl status sshd --no-pager
```

```

-bash-5.3# systemctl daemon-reload
-bash-5.3# systemctl enable sshd.service
Created symlink '/etc/systemd/system/multi-user.target.wants/sshd.service' -> '/etc/systemd/system/sshd.service'.
-bash-5.3# systemctl start sshd.service
-bash-5.3# systemctl status sshd.service
• sshd.service - OpenSSH Daemon
   Loaded: loaded (/etc/systemd/system/sshd.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-01-08 18:30:26 -03; 9s ago
  Invocation: 3c88b244c03342c7a3726c59396d2533
    Main PID: 21566 (sshd)
      Tasks: 1 (limit: 9534)
     Memory: 1.4M (peak: 1.5M)
        CPU: 30ms
    CGroup: /system.slice/sshd.service
            └─21566 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jan 08 18:30:26 lfs systemd[1]: Starting OpenSSH Daemon...
Jan 08 18:30:26 lfs sshd[21566]: Server listening on 0.0.0.0 port 22.
Jan 08 18:30:26 lfs sshd[21566]: Server listening on :: port 22.
Jan 08 18:30:26 lfs systemd[1]: Started OpenSSH Daemon.
-bash-5.3# systemctl enable sshd.service

```

i) Configuramos el archivo y verificamos el funcionamiento

```
cat >> /etc/ssh/sshd_config << "EOF"
```

```
Port 22
```

```
PermitRootLogin yes
```

```
PasswordAuthentication yes
```

PubkeyAuthentication yes

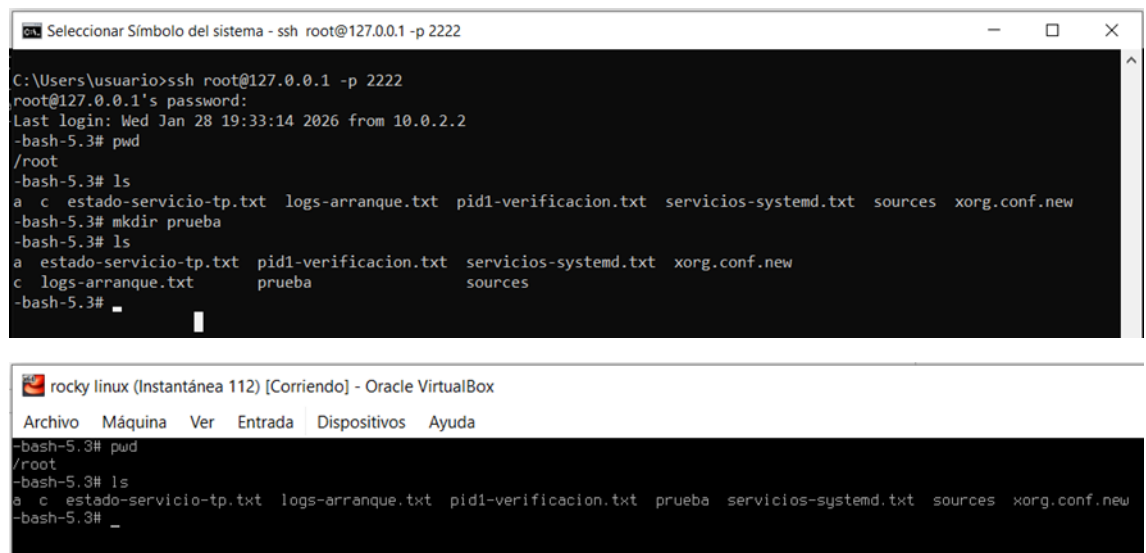
EOF

sshd -t -f /etc/ssh/sshd\_config

systemctl restart sshd

```
-bash-5.3# cat >> /etc/ssh/sshd_config << "EOF"
> Port 22
> PermitRootLogin yes
> PasswordAuthentication yes
> PubkeyAuthentication yes
> EOF
-bash-5.3# sshd -t -f /etc/ssh/sshd_config
-bash-5.3# systemctl restart sshd
-bash-5.3# ss -tulpn | grep ':22'
tcp    LISTEN 0      0            0.0.0.0:22      0.0.0.0:*      users:((("sshd",p
id=59140,fd=6))
tcp    LISTEN 0      0            *:22          *:.*          users:((("sshd",p
id=59140,fd=7))
-bash-5.3#
```

j) Prueba con WindowsPowerShell



The first screenshot shows a Windows PowerShell terminal window titled "Seleccionar Símbolo del sistema - ssh root@127.0.0.1 -p 2222". The user connects to a Rocky Linux system. The terminal shows the password prompt, the last login time (Wed Jan 28 19:33:14 2026 from 10.0.2.2), and the user's current directory (/root). The user then runs 'ls' and 'mkdir prueba', showing the directory contents before and after the creation of the 'prueba' directory.

The second screenshot shows a terminal window titled "rocky linux (Instantánea 112) [Corriendo] - Oracle VirtualBox". It displays the same sequence of commands and output as the first screenshot, confirming the successful setup of the SSH environment.



## Interfaz Gráfica Ligera

### Capítulo 24 – Graphical Environments

Instalación de **wget-1.25** (recomendado por el manual BLFS)

12/01/2026

- Preparación del **Xorg environment**.
- Instalación de **util-macros1.20.2** a las 15:00.
  - Contiene la m4 macros utilizada por todos los paquetes Xorg.
  - Necesario para el Xorg build environment y para xorgproto-2024.1.
- Instalación de **xorgproto-2024.1** a las 15:30.
  - Proporciona los header files necesarios para construir el sistema X Window y para permitir que otras aplicaciones se compilen para el sistema X Window instalado.
  - Necesario para libXau-1.0.12 y para libXdmcp-1.1.5.
  -

Notes: no notes.

- Instalación de **libXau-1.0.12** a las 16:00
  - El paquete libXau contiene una biblioteca que implementa el Protocolo de Autorización X11.
  - Necesario para libxcb-1.17.0

Notes: no notes

- Instalación de **libXdmcp-1.1.5** a las 16:52
  - El paquete libXdmcp contiene una biblioteca que implementa el Protocolo de Control del Gestor de Pantallas X (X Display Manager Control Protocol). Esto es útil para permitir que los clientes interactúen con el X Display Manager.
  - Necesario para Mesa-25.1.8
  -

Notes: make check succesful

- Instalación de **xcb-proto-1.17.0** a las 17:01
  - El paquete xcb-proto proporciona las descripciones del protocolo XML-XCB que libxcb utiliza para generar la mayor parte de su código y su API.
  - Recomendado para el Xorg build environment
  - Necesario para libxcb-1.17.0
  -

Notes: make check succesful

- Instalación de **libxcb-1.17.0** a las 17:20
  - El paquete libxcb proporciona una interfaz para el protocolo del Sistema de Ventanas X, la cual reemplaza a la interfaz actual Xlib. Además, Xlib también puede usar XCB como capa de transporte, lo que permite que el software realice solicitudes y reciba respuestas utilizando ambas.
  - Necesario para Xorg libraries y para xcb-util-0.4.1

○

Notes: make check succesfuk

- Instalación de **Fontconfig-2.17.1** a las 17:00
  - El paquete Fontconfig contiene una biblioteca y programas de soporte utilizados para configurar y personalizar el acceso a las fuentes.
  - Necesario para Xorg Libraries
  -

Notes: no notes

- Instalación de **Xorg Libraries** a las 19:00
  - Las bibliotecas de Xorg proporcionan rutinas de biblioteca que se utilizan dentro de todas las aplicaciones del Sistema de Ventanas X.
  - Necesario para Mesa-25.1.8
  - Librerías instaladas:
    - xtrans 1.6.0
    - libX11 1.8.12
    - libXext 1.3.6
    - libFS 1.0.10
    - libICE 1.1.2
    - libSM 1.2.6
    - libXScrnSaver 1.2.4
    - libXt 1.3.1
    - libXmu 1.2.1
    - libXpm 3.5.17
    - libXaw 1.0.16
    - libXfixes 6.0.1
    - libXcomposite 0.4.6
    - libXrender 0.9.12
    - libXcursor 1.2.3
    - libXdamage 1.1.6
    - libfontenc 1.1.8
    - libXfont2 2.0.7
    - libXft 2.3.9
    - libXi 1.8.2
    - libXinerama 1.1.5
    - libXrandr 1.5.4
    - libXres 1.2.2
    - libXtst 1.2.5
    - libXv 1.0.13
    - libXvMC 1.0.14
    - libXxf86dga 1.1.6
    - libXxf86vm 1.1.6
    - libpciaccess 0.18.1
    - libxkbfile 1.1.3
    - libxshmfence 1.3.3
    - libXpresent 1.0.1

Notes: ningún make check hecho. Tuve que instalar freetype

- Instalación de **libxcvt-0.1.3** a las 20:00
  - La librería libxcvt proporciona una versión independiente de la implementación del servidor X del generador de modelinas de temporización estándar VESA CVT. Está diseñada para ser un reemplazo directo de la versión que antes suministraba el servidor Xorg.
  - Necesario para XCB Utilities
  -

Notes: no notes.

13/01/2026

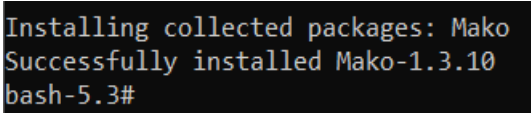
- Instalación de **xcb-util-0.4.1**
  - El paquete xcb-util proporciona extensiones adicionales a la biblioteca XCB, muchas de las cuales antes se encontraban en Xlib, pero que no forman parte del protocolo X principal.
  - Necesario para XCB Utilities
  -

Notes: no notes.

- Instalación de **XCB Utilities**
  - Las XCB Utilities proporcionan extensiones que los desarrolladores pueden usar al crear software para el sistema de ventanas X. Se acaba de instalar xcb-util-0.4.1, pero estas utilidades ofrecen aún más extensiones de las que el software de X Window puede depender.
  -

Notes: no notes.

14/01/2026

- Instalación de **Mako-1.3.10** a las 12:48
  - Mako es un módulo de Python que implementa un sistema de plantillas ultrarrápido y liviano para la plataforma Python.
  - Necesario para Mesa-25.1.8
  - Installing collected packages: Mako  
Successfully installed Mako-1.3.10  
bash-5.3#

Notes: no notes.

- Instalación de **libyaml-0.2.5** a las 12:49
  - El paquete yaml contiene una biblioteca en C para analizar y generar código YAML (YAML Ain't Markup Language).
  - Necesario para PyYAML-6.0.2

```
Testsuite summary for yaml 0.2.5
```

```
# TOTAL: 2  
# PASS: 2  
# SKIP: 0  
# XFAIL: 0  
# FAIL: 0  
# XPASS: 0  
# ERROR: 0
```

○

Notes: no notes.

- Instalación de **cython-3.1.3** a las 12:50

- El paquete Cython proporciona un compilador para escribir extensiones en C para el lenguaje Python.
- Necesario para PyYAML-6.0.2

```
Installing collected packages: Cython  
Successfully installed Cython-3.1.3  
bash-5.3# cd ..
```

○

Notes: no notes.

- Instalación de **PyYAML-6.0.2** a las 12:55

- PyYAML es un módulo de Python que implementa el parser y el emisor de YAML de próxima generación.
- Necesario para Mesa-25. 1.8

```
Installing collected packages: PyYAML  
Successfully installed PyYAML-6.0.2  
bash-5.3#
```

○

Notes: no notes.

- Instalación de **lspci** a las 13:16

- El paquete PCI Utils contiene un conjunto de programas para listar dispositivos PCI, inspeccionar su estado y configurar sus registros de configuración.
- Instalado para conocer los controllers compatibles con la VM

```
bash-5.3# lspci | grep VGA  
00:02.0 VGA compatible controller: VMware SVGA II Adapter
```

○

Notes: no notes.

- Instalación de **libgpg-error-1.55** a las 13:24

- El paquete libgpg-error contiene una biblioteca que define valores de error comunes para todos los componentes de GnuPG.
- Necesario para libgcrypt

```

echo=1
PASS: t-argparse
PASS: t-logging
PASS: t-stringutils
PASS: t-malloc
PASS: t-spawn
PASS: t-strlist
PASS: t-name-value
PASS: t-lock-single-posix
=====
All 15 tests passed
=====

```

○

Notes: no notes.

- Instalación de **libgcrypt-1.11.2** a las 13:38

- El paquete libgcrypt contiene una biblioteca criptográfica de propósito general basada en el código utilizado en GnuPG. La biblioteca proporciona una interfaz de alto nivel para bloques de construcción criptográficos mediante una API extensible y flexible.

○

```

=====
All 38 tests passed
(2 tests were not run)
=====

```

○

Notes: no notes.

- Instalación de **libunwind-1.8.2** a las 14:12

- El paquete libunwind contiene una interfaz de programación en C (API) portátil y eficiente para determinar la cadena de llamadas de un programa. Además, la API proporciona los medios para manipular el estado preservado (guardado por la función llamada) de cada marco de llamada y para reanudar la ejecución en cualquier punto de la cadena de llamadas (salto no local). La API admite operaciones tanto locales (en el mismo proceso) como remotas (entre procesos).

○

```

=====
Testsuite summary for libunwind 1.8.2
=====
# TOTAL: 40
# PASS: 38
# SKIP: 2
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====

```

○

Notes: no notes.

- Instalación de **nettle-3.10.2** a las 14:28

- El paquete Nettle contiene una biblioteca criptográfica de bajo nivel diseñada para integrarse fácilmente en muchos contextos.

○

```

=====
All 117 tests passed
=====
make[1]: Leaving directory '/root/sources/nettle-3.10.2/testsuite'
Making check in examples
make[1]: Entering directory '/root/sources/nettle-3.10.2/examples'
TEST_SHLIB_DIR="/root/sources/nettle-3.10.2/.lib" \
  srcdir="." EMULATOR="" EXEEXT="" \
  "../run-tests rsa-sign-test rsa-verify-test rsa-encrypt-test
xxxxxx
xxxxxx
PASS: rsa-sign
PASS: rsa-verify
PASS: rsa-encrypt
=====
All 3 tests passed
=====
make[1]: Leaving directory '/root/sources/nettle-3.10.2/examples'
bash-5.3#

```

○

Notes: no notes.

- Instalación de **six-1.17.0** a las 14:44
  - Six es una biblioteca de compatibilidad que permite escribir código válido tanto para Python 2 como para Python 3.

○

```

Installing collected packages: six
Successfully installed six-1.17.0
bash-5.3#

```

○

Notes: no notes.

- Instalación de **libaio-0.3.113** a las 16:25
  - El paquete libaio es una facilidad de E/S asíncrona ("async I/O" o "aio") que cuenta con una API y un conjunto de capacidades más ricos que la sencilla E/S asíncrona POSIX. Esta biblioteca, libaio, proporciona la API nativa de Linux para E/S asíncrona. La facilidad de E/S asíncrona POSIX requiere esta biblioteca para ofrecer capacidades de E/S asíncrona aceleradas por el kernel, al igual que las aplicaciones que necesitan la API nativa de E/S asíncrona de Linux.

○

```

Pass: 19 Fail: 0 Skip: 0
Test run complete at Wed Jan 14 04:25:22 PM -03 2026
make[1]: Leaving directory '/root/sources/libaio-0.3.113/harness'
bash-5.3#

```

○

Notes: no notes.

- Instalación de **icu-77.1** a las 22:26
  - El paquete International Components for Unicode (ICU) es un conjunto maduro y ampliamente utilizado de bibliotecas en C/C++ que proporcionan soporte para Unicode y para la internacionalización de aplicaciones de software. ICU es altamente portable y ofrece a las aplicaciones los mismos resultados en todas las plataformas.

○

```
ALL TESTS SUMMARY:  
All tests OK: testdata intltest iotest cintltst
```

○

Notes: no notes.

- Instalación de **libxml2** a las 22:38

- El paquete libxml2 contiene bibliotecas y utilidades utilizadas para analizar archivos XML.

○

```
bash-5.3# grep -E '^Total|expected|Ran' check.log  
Total 3943 tests, no errors  
Total 49 tests, no errors  
Total: 976 functions, 323618 tests, 0 errors  
Ran 2273 tests, 15 errors, 0 leaks  
Total 2273 tests, 15 errors, 0 leaks  
15 errors were expected  
Ran 1035 tests, 3 errors, 0 leaks  
3 errors were expected  
Ran 253 tests, no errors  
Ran 183 tests, no errors  
Ran 0 tests (0 schemata), no errors  
Ran 0 tests (0 schemata), no errors  
Ran 0 tests (0 schemata), no errors  
Total 1471 tests, no errors  
bash-5.3#
```

○

Notes: no notes.

- Instalación de **lzo-2.10** a las 22:49

- LZO es una biblioteca de compresión de datos adecuada para la descompresión y compresión en tiempo real. Esto significa que prioriza la velocidad sobre la tasa de compresión.

○

```
lzotest: execution time: 0 seconds  
  
===== All checks passed. Type 'make test' for more exhaustive testing. =====  
  
make[1]: Leaving directory '/root/sources/lzo-2.10'  
bash-5.3#
```

○

```
LZO real-time data compression library (v2.10, Mar 01 2017).  
Copyright (C) 1996-2017 Markus Franz Xavier Johannes Oberhumer  
All Rights Reserved.
```

```
compressed 131072 bytes into 593 bytes  
decompressed 593 bytes back into 131072 bytes
```

```
miniLZO simple compression test passed.
```

```
===== All tests passed. Now you are ready to install LZO. =====
```

```
bash-5.3#
```

○

Notes: no notes.

- Instalación de **libarchive-3.8.1** finalizada a las 23:13
  - La biblioteca libarchive proporciona una única interfaz para leer y escribir diversos formatos de compresión.

○

```
=====
Testsuite summary for libarchive 3.8.1
=====
# TOTAL: 5
# PASS: 5
# SKIP: 0
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====
```

○

Notes: no notes.

- Instalación de **docbook-xml-4.5** a las 23:36
  - El paquete DocBook-4.5 XML DTD-4.5 contiene definiciones de tipo de documento para la verificación de archivos de datos XML según el conjunto de reglas de DocBook. Son útiles para estructurar libros y documentación de software siguiendo un estándar, lo que permite aprovechar transformaciones ya escritas para ese estándar.

○

```
> xmlcatalog --noout --add "delegatePublic" \
>   "-//OASIS//ENTITIES DocBook XML" \
>   "file:///etc/xml/docbook" \
>   /etc/xml/catalog &&
public">
> xmlcatalog --noout --add "delegatePublic" \
>   "-//OASIS//DTD DocBook XML" \
>   "file:///etc/xml/docbook" \
>   /etc/xml/catalog &&
>
> xmlcatalog --noout --add "delegateSystem" \
>   "http://www.oasis-open.org/docbook/" \
>   "file:///etc/xml/docbook" \
>   /etc/xml/catalog &&
I" >
> xmlcatalog --noout --add "delegateURI" \
>   "http://www.oasis-open.org/docbook/" \
>   "file:///etc/xml/docbook" \
>   /etc/xml/catalog
bash-5.3#
```

○

Notes: no notes.



15/01/2026

- Instalación de **Valgrind-3.25.1** hecho a las 01:10
  - Valgrind is an instrumentation framework for building dynamic analysis tools. There are Valgrind tools that can automatically detect many memory management and threading bugs, and profile programs in detail. Valgrind can also be used to build new tools.
  - Opcional para Libdrm-2.4.125.

○

```
== 844 tests, 4 stderr failures, 0 stdout failures, 1 stderrB failure, 2 stdoutB failures, 0 post failures ==
gdbserver_tests/mcinfcallWSRU          (stderrB)
gdbserver_tests/nlcontrolc             (stdoutB)
gdbserver_tests/nlvgdbsigqueue         (stdoutB)
memcheck/tests/dscr_belowsp           (stderr)
helgrind/tests/getaddrinfo             (stderr)
helgrind/tests/tc22_exit_w_lock        (stderr)
drd/tests/std_thread2                  (stderr)

...checking makefile consistency
...checking header files and include directives
make: *** [Makefile:1457: regtest] Error 1
bash-5.3#
```

○

Notes: no notes.

- Instalación de **libunistring-1.3** a las 01:39
  - libunistring es una biblioteca que proporciona funciones para manipular cadenas Unicode y para manipular cadenas en C según el estándar Unicode.

○

```
=====  
Testsuite summary for  
=====
```

```
# TOTAL: 665  
# PASS: 629  
# SKIP: 36  
# XFAIL: 0  
# FAIL: 0  
# XPASS: 0  
# ERROR: 0  
=====
```

○

Notes: no notes.

- Instalación de **libidn2-2.3.8** a las 02:21
  - libidn2 es un paquete diseñado para el manejo de cadenas internacionalizadas, basado en los estándares del grupo de trabajo IDN del Internet Engineering Task Force (IETF), y está pensado para nombres de dominio internacionalizados.

○

```
=====
Testsuite summary for Libidn2 2.3.8
=====
```

```
# TOTAL: 12
# PASS: 12
# SKIP: 0
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====
```

○

Notes: no notes.

- Instalación de **libpsl-0.21.5** finalizada a las 02:24

- El paquete libpsl proporciona una biblioteca para acceder y resolver información de la Public Suffix List (PSL). La PSL es un conjunto de nombres de dominio más allá de los sufijos estándar, como .com.

○

```
bash-5.3# ninja test
[21/22] Running all tests
1/8 test-is-public OK 0.05s
2/8 test-is-cookie-domain-acceptable OK 0.05s
3/8 test-is-public-builtin OK 0.01s
4/8 test-registrable-domain OK 0.02s
5/8 libpsl_idn2_fuzzer OK 0.04s
6/8 test-is-public-all OK 0.58s
7/8 libpsl_idn2_load_dafsa_fuzzer OK 0.03s
8/8 libpsl_idn2_load_fuzzer OK 0.68s

Ok: 8
Fail: 0

Full log written to /root/sources/libpsl-0.21.5/build/meson-logs/testlog.txt
bash-5.3# ninja install
```

○

Notes: no notes.

- Instalación de **libtasn1-4.20.0** finalizada a las 10:48

- libtasn1 es una biblioteca en C altamente portable que codifica y decodifica datos DER/BER siguiendo un esquema ASN.1.

○

```
=====
Testsuite summary for GNU Libtasn1 4.20.0
=====
```

```
# TOTAL: 31
# PASS: 31
# SKIP: 0
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====
```

○

Notes: no notes.

- Instalación de **p11-kit-0.25.5** finalizada a las 10:59

- El paquete p11-kit proporciona una forma de cargar y enumerar módulos PKCS #11 (un estándar de interfaz de tokens criptográficos).

○

```
Ok:                67
Fail:              0

Full log written to /root/sources/p11-kit-0.25.5/p11-build/meson-logs/testlog.txt
bash-5.3#
```

○

Notes: no notes.

- Instalación de **curl-8.15.0** finalizada a las 12:53

- El paquete cURL contiene una utilidad y una biblioteca utilizadas para transferir archivos mediante la sintaxis de URL a cualquiera de los siguientes protocolos: DICT, FILE, FTP, FTPS, GOPHER, GOPHERS, HTTP, HTTPS, IMAP, IMAPS, LDAP, LDAPS, MQTT, POP3, POP3S, RTSP, SMB, SMBS, SMTP, SMPTS, TELNET y TFTP. Su capacidad para descargar y subir archivos puede incorporarse en otros programas para soportar funciones como la transmisión de medios.

○

```
bash-5.3# bash-5.3# curl --version
curl 8.15.0 (x86_64-pc-linux-gnu) libcurl/8.15.0 OpenSSL/3.5.4 zlib/1.3.1 zstd/1.5.7 libidn2/2.3.8 libpsl/0.21.5
Release-Date: 2025-07-16
Protocols: dict file ftp ftps gopher gophers http https imap imaps ipfs ipns mqtt pop3 pop3s rtsp smb smbs smtp smtps telnet tftp ws wss
Features: alt-svc AsynchDNS HSTS HTTPS-proxy IDN IPv6 Largefile libz NTLM PSL SSL threadsafe TLS-SRP UnixSockets zstd
bash-5.3# curl https://example.com
<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.<p><a href="https://iana.org/domains/example">Learn more</a></div></body></html>
bash-5.3# curl -Is https://www.google.com
HTTP/1.1 200 OK
```

○

Notes: no notes.

- Instalación de **libuv-v1.51.0** finalizada a las 13:18

- El paquete libuv es una biblioteca de soporte multiplataforma con un enfoque en E/S asíncrona.

○

```

bash-5.3# make install
make[1]: Entering directory '/root/sources/libuv-v1.51.0'
/usr/bin/mkdir -p '/usr/lib'
/bin/sh ./libtool --mode=install /usr/bin/install -c libuv.la '/usr/lib'
libtool: install: /usr/bin/install -c .libs/libuv.so.1.0.0 /usr/lib/libuv.so.1.0.0
libtool: install: (cd /usr/lib && { ln -s -f libuv.so.1.0.0 libuv.so.1 || { rm -f libuv.so.1 && ln -s li
buv.so.1.0.0 libuv.so.1; }; })
libtool: install: (cd /usr/lib && { ln -s -f libuv.so.1.0.0 libuv.so || { rm -f libuv.so && ln -s libuv.
so.1.0.0 libuv.so; }; })
libtool: install: /usr/bin/install -c .libs/libuv.lai /usr/lib/libuv.la
libtool: finish: PATH="/usr/bin:/bin:/usr/sbin:/sbin:/sbin" ldconfig -n /usr/lib

-----
Libraries have been installed in:
  /usr/lib

If you ever happen to want to link against installed libraries
in a given directory, LIBDIR, you must either use libtool, and
specify the full pathname of the library, or use the '-LLIBDIR'
flag during linking and do at least one of the following:
- add LIBDIR to the 'LD_LIBRARY_PATH' environment variable
  during execution
- add LIBDIR to the 'LD_RUN_PATH' environment variable
  during linking
- use the '-Wl,-rpath -Wl,LIBDIR' linker flag
- have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.
-----
/usr/bin/mkdir -p '/usr/include'
/usr/bin/install -c -m 644 include/uv.h '/usr/include'
/usr/bin/mkdir -p '/usr/lib/pkgconfig'
/usr/bin/install -c -m 644 libuv.pc '/usr/lib/pkgconfig'
/usr/bin/mkdir -p '/usr/include/uv'
/usr/bin/install -c -m 644 include/uv/errno.h include/uv/threadpool.h include/uv/version.h include/uv/u
nix.h include/uv/linux.h '/usr/include/uv'
make[1]: Leaving directory '/root/sources/libuv-v1.51.0'
bash-5.3#
○ bash-5.3# pkg-config --modversion libuv
1.51.0
○ bash-5.3#

```

Notes: no notes.

- Instalación de **nghttp2-1.66.0** finalizada a las 13:23
  - nghttp2 es una implementación de HTTP/2 y de su algoritmo de compresión de cabeceras, HPACK.
  -

```

PASS: main
PASS: failmalloc
=====
Testsuite summary for nghttp2 1.66.0
=====
# TOTAL: 2
# PASS: 2
# SKIP: 0
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====
○

```

Notes: no notes.

- Instalación de **cmake-4.1.0** finalizada a las 16:37
  - El paquete CMake contiene un conjunto de herramientas moderno utilizado para generar Makefiles. Es el sucesor del script de configuración auto-generado y tiene

como objetivo ser independiente de la plataforma y del compilador. Un usuario importante de CMake es KDE desde la versión 4.

○

```
[ 97%] Linking CXX executable ../../bin/ctresalloc
[ 97%] Built target ctresalloc
[ 97%] Building C object Tests/RunCMake/CMakeFiles/pseudo_llvm-rc.dir/pseudo_llvm-rc.c.o
[ 97%] Linking C executable pseudo_llvm-rc
[ 97%] Built target pseudo_llvm-rc
[ 97%] Building C object Tests/RunCMake/CMakeFiles/print_stdin.dir/print_stdin.c.o
[ 97%] Linking C executable print_stdin
[ 97%] Built target print_stdin
[ 97%] Building C object Tests/RunCMake/CMakeFiles/pseudo_emulator.dir/pseudo_emulator.c.o
[ 97%] Linking C executable pseudo_emulator
[ 97%] Built target pseudo_emulator
[ 97%] Building C object Tests/RunCMake/CMakeFiles/pseudo_emulator_custom_command.dir/pseudo_emulator_custom_command.c.o
[ 97%] Linking C executable pseudo_emulator_custom_command
[ 97%] Built target pseudo_emulator_custom_command
[ 97%] Building C object Tests/RunCMake/CMakeFiles/pseudo_emulator_custom_command_arg.dir/pseudo_emulator_custom_command_arg.c.o
[ 98%] Linking C executable pseudo_emulator_custom_command_arg
[ 98%] Built target pseudo_emulator_custom_command_arg
[ 99%] Building C object Tests/RunCMake/CMakeFiles/pseudo_tidy.dir/pseudo_tidy.c.o
[ 99%] Linking C executable pseudo_tidy
[ 99%] Built target pseudo_tidy
[ 99%] Building C object Tests/RunCMake/CMakeFiles/pseudo_iwyu.dir/pseudo_iwyu.c.o
[ 99%] Linking C executable pseudo_iwyu
[ 99%] Built target pseudo_iwyu
[ 99%] Building C object Tests/RunCMake/CMakeFiles/pseudo_cpplint.dir/pseudo_cpplint.c.o
[ 99%] Linking C executable pseudo_cpplint
[ 99%] Built target pseudo_cpplint
[ 99%] Building C object Tests/RunCMake/CMakeFiles/pseudo_cppcheck.dir/pseudo_cppcheck.c.o
[ 99%] Linking C executable pseudo_cppcheck
[ 99%] Built target pseudo_cppcheck
[ 99%] Building CXX object Tests/FindPackageModeMakefileTest/CMakeFiles/foo.dir/foo.cpp.o
[100%] Linking CXX static library libfoo.a
[100%] Built target foo
bash-5.3#
```

○

Notes: no notes.

16/01/2026

- Instalación de **LLVM-20.1.8** finalizada a las 09:22
  - El paquete LLVM contiene una colección de tecnologías modulares y reutilizables para compiladores y cadenas de herramientas. Las bibliotecas principales de LLVM (Low Level Virtual Machine Core) proporcionan un optimizador moderno independiente del origen y del destino, junto con soporte de generación de código para muchos procesadores populares (¡así como algunos menos comunes!). Estas bibliotecas se construyen alrededor de una representación de código bien especificada conocida como representación intermedia de LLVM ("LLVM IR").
  - Clang proporciona nuevos front-ends para C, C++, Objective C y Objective C++ para LLVM, y es requerido por algunos paquetes de escritorio como Firefox y por Rust si se construye utilizando LLVM del sistema.
  - El paquete Compiler RT proporciona bibliotecas de tiempo de ejecución para sanitización y perfilado para desarrolladores que usan Clang y LLVM.
  - Necesario para llvmpipe
  -

```
[5170/5170] Linking CXX shared module unittests/Passes/Plugins/DoublerPlugin.so
bash-5.3#
```

○

Notes: no notes.

17/01/2026

- Instalación de **Mesa-25.1.8** finalizada a las 18:23
  - Mesa es una biblioteca de gráficos 3D compatible con OpenGL.
  - Depende de Xorg Libraries, libdrm-2.4.125, Mako-1.3.10 y PyYAML-6.0.2
  - Parche recomendado: mesa-add\_xdemos-4.patch.
  - Built with: mesa-add\_xdemos-4.patch, Xorg Libraries, libdrm-2.4.125, Mako-1.3.10, PyYAML-6.0.2, LLVM-20.1.8, libcrypt-1.11.2, libunwind-1.8.2, Nettle-3.10.2, Valgrind-3.25.1 y mesa-demos.

○

```
Ok:                48
Fail:              0

Full log written to /root/sources/mesa-25.1.8/build/meson-logs/testlog.txt
-bash-5.3#
```

○

Notes: no notes.

18/01/2026

- Instalación de **libpng-1.6.50** finalizada a las 12:25
  - El paquete libpng contiene bibliotecas utilizadas por otros programas para leer y escribir archivos PNG. El formato PNG fue diseñado como reemplazo de GIF y, en menor medida, de TIFF, ofreciendo muchas mejoras y extensiones, además de no presentar problemas de patentes.

○

```
=====
Testsuite summary for libpng 1.6.50
=====
# TOTAL: 32
# PASS:  32
# SKIP:  0
# XFAIL: 0
# FAIL:  0
# XPASS: 0
# ERROR: 0
=====
```

○

Notes: no notes.

- Instalación de **pixman-0.46.4** finalizada a las 18:37
  - El paquete Pixman contiene una biblioteca que proporciona funciones de manipulación de píxeles a bajo nivel, como composición de imágenes y rasterización de trapecios.

○

```

[0/1] Running all tests
1/35 oob-test OK 0.03s
2/35 infinite-loop OK 0.03s
3/35 trap-crasher OK 0.03s
4/35 region-translate-test OK 0.02s
5/35 fetch-test OK 0.02s
6/35 fence-image-self-test OK 0.08s
7/35 a1-trap-test OK 0.03s
8/35 prng-test OK 0.03s
9/35 radial-invalid OK 0.03s
10/35 pdf-op-test OK 0.05s
11/35 region-test OK 0.09s
12/35 combiner-test OK 0.03s
13/35 region-fractional-test OK 0.11s
14/35 alpha-loop OK 0.06s
15/35 scaling-crash-test OK 0.53s
16/35 scaling-helpers-test OK 1.27s
17/35 rotate-test OK 1.07s
18/35 gradient-crash-test OK 0.96s
19/35 pixel-test OK 4.24s
20/35 matrix-test OK 8.43s
21/35 alphamap OK 19.90s
22/35 filter-reduction-test OK 20.03s
23/35 composite-traps-test OK 18.88s
24/35 glyph-test OK 32.54s
25/35 region-contains-test OK 65.24s
26/35 solid-test OK 40.52s
27/35 stress-test OK 98.11s
28/35 cover-test OK 106.34s
29/35 blitters-test OK 79.51s
30/35 affine-test OK 107.87s
scaling-test time out (After 120 seconds)
31/35 scaling-test TIMEOUT 120.03s killed by signal 15 SIGTERM
>>> UBSAN_OPTIONS=halt_on_error=1:abort_on_error=1:print_summary=1:print_stacktrace=1 MSAN_OPTIONS=halt_on_error=1:abort_on_error=1:print_summary=1:print_stacktrace=1 LD_LIBRARY_PATH=/root/sources/pixman-0.46.4/build/pixman ASAN_OPTIONS=halt_on_error=1:abort_on_error=1:print_summary=1 MESON_TEST_ITERATION=1 MALLOC_PERTURB=169 /root/sources/pixman-0.46.4/build/test/scaling-test

composite time out (After 120 seconds)
32/35 composite TIMEOUT 120.03s killed by signal 15 SIGTERM
>>> UBSAN_OPTIONS=halt_on_error=1:abort_on_error=1:print_summary=1:print_stacktrace=1 MALLOC_PERTURB=90 MSAN_OPTIONS=halt_on_error=1:abort_on_error=1:print_summary=1:print_stacktrace=1 LD_LIBRARY_PATH=/root/sources/pixman-0.46.4/build/pixman ASAN_OPTIONS=halt_on_error=1:abort_on_error=1:print_summary=1 MESON_TEST_ITERATION=1 /root/sources/pixman-0.46.4/build/test/composite

33/35 neg-stride-test OK 0.03s
34/35 thread-test OK 0.97s
tolerance-test time out (After 120 seconds)
35/35 tolerance-test TIMEOUT 120.03s killed by signal 15 SIGTERM
-----
35/35 tolerance-test TIMEOUT 120.03s killed by signal 15 SIGTERM
>>> UBSAN_OPTIONS=halt_on_error=1:abort_on_error=1:print_summary=1:print_stacktrace=1 MSAN_OPTIONS=halt_on_error=1:abort_on_error=1:print_summary=1:print_stacktrace=1 LD_LIBRARY_PATH=/root/sources/pixman-0.46.4/build/pixman ASAN_OPTIONS=halt_on_error=1:abort_on_error=1:print_summary=1 MESON_TEST_ITERATION=1 MALLOC_PERTURB=38 /root/sources/pixman-0.46.4/build/test/tolerance-test

Summary of Failures:

31/35 scaling-test TIMEOUT 120.03s killed by signal 15 SIGTERM
32/35 composite TIMEOUT 120.03s killed by signal 15 SIGTERM
35/35 tolerance-test TIMEOUT 120.03s killed by signal 15 SIGTERM

Ok: 32
Fail: 0
Timeout: 3

Full log written to /root/sources/pixman-0.46.4/build/meson-logs/testlog.txt
FAILED: [code=3] meson-internal_test
/usr/bin/meson test --no-rebuild --print-errorlogs
ninja: build stopped: subcommand failed.
-bash-5.3#

```

Notes: no notes.

19/01/2026

- Instalación de **libepoxy-1.5.10** finalizada a las 3:50
  - libepoxy es una biblioteca para el manejo de punteros a funciones de OpenGL.
  -

```
-bash-5.3# ninja test
[0/1] Running all tests
1/17 header_guards          OK          0.02s
2/17 misc_defines           OK          0.02s
3/17 khronos_typedefs       OK          0.02s
4/17 gl_version             OK          0.03s
5/17 egl_has_extension_nocontext SKIP        0.03s  exit status 77
6/17 egl_epoxy_api          SKIP        0.03s  exit status 77
7/17 egl_gles1_without_glx  SKIP        0.03s  exit status 77
8/17 egl_gles2_without_glx  SKIP        0.03s  exit status 77
9/17 egl_gl                 SKIP        0.29s  exit status 77
10/17 glx_beginend          SKIP        0.29s  exit status 77
11/17 glx_public_api        SKIP        0.03s  exit status 77
12/17 glx_public_api_core   SKIP        0.03s  exit status 77
13/17 glx_glxgetprocaddress_nocontext SKIP        0.03s  exit status 77
14/17 glx_has_extension_nocontext SKIP        0.03s  exit status 77
15/17 glx_shared_znow       SKIP        0.03s  exit status 77
16/17 glx_alias_prefer_same_name SKIP        0.03s  exit status 77
17/17 glx_gles2            SKIP        0.02s  exit status 77

Ok:          4
Fail:        0
Skipped:     13

Full log written to /root/sources/libepoxy-1.5.10/build/meson-logs/testlog.txt
-bash-5.3#
```



○

```
libepoxy 1.5.10

User defined options
  buildtype: release
  prefix   : /usr

Found ninja-1.13.1 at /usr/bin/ninja
[55/55] Linking target test/glx_beginend
-bash-5.3# ninja test
[0/1] Running all tests
1/17 header_guards          OK          0.02s
2/17 misc_defines           OK          0.02s
3/17 khronos_typedefs       OK          0.02s
4/17 gl_version             OK          0.03s
5/17 egl_has_extension_nocontext SKIP       0.03s  exit status 77
6/17 egl_epoxy_api          SKIP       0.03s  exit status 77
7/17 egl_gles1_without_glx  SKIP       0.03s  exit status 77
8/17 egl_gles2_without_glx  SKIP       0.03s  exit status 77
9/17 egl_gl                 SKIP       0.29s  exit status 77
10/17 glx_beginend          SKIP       0.29s  exit status 77
11/17 glx_public_api        SKIP       0.03s  exit status 77
12/17 glx_public_api_core   SKIP       0.03s  exit status 77
13/17 glx_glxgetprocaddress_nocontext SKIP       0.03s  exit status 77
14/17 glx_has_extension_nocontext SKIP       0.03s  exit status 77
15/17 glx_shared_zenow      SKIP       0.03s  exit status 77
16/17 glx_alias_prefer_same_name SKIP       0.03s  exit status 77
17/17 glx_gles2             SKIP       0.02s  exit status 77

Ok:          4
Fail:        0
Skipped:     13

Full log written to /root/sources/libepoxy-1.5.10/build/meson-logs/testlog.txt
-bash-5.3# ninja install
[0/1] Installing files
Installing include/epoxy/gl_generated.h to /usr/include/epoxy
Installing include/epoxy/egl_generated.h to /usr/include/epoxy
Installing include/epoxy/glx_generated.h to /usr/include/epoxy
Installing src/libepoxy.so.0.0.0 to /usr/lib64
Installing /root/sources/libepoxy-1.5.10/include/epoxy/common.h to /usr/include/epoxy
Installing /root/sources/libepoxy-1.5.10/include/epoxy/gl.h to /usr/include/epoxy
Installing /root/sources/libepoxy-1.5.10/include/epoxy/egl.h to /usr/include/epoxy
Installing /root/sources/libepoxy-1.5.10/include/epoxy/glx.h to /usr/include/epoxy
Installing /root/sources/libepoxy-1.5.10/build/meson-private/epoxy.pc to /usr/lib64/pkgconfig
Installing symlink pointing to libepoxy.so.0.0.0 to /usr/lib64/libepoxy.so.0
Installing symlink pointing to libepoxy.so.0 to /usr/lib64/libepoxy.so
-bash-5.3#
```

Notes: no notes.

- Instalación de **xorg-server-21.1.18** finalizada a las
  - El Xorg Server es el núcleo del X Window system.
  - Crucial para el sistema X Window.
  - Necesario para Xorg Synaptics Driver-1.10.0, twm-1.0.13.1

```

Configuring xorg-server.pc using configuration
Build targets in project: 66
WARNING: Project specifies a minimum meson_version '>= 0.47.0' but uses features which were added in newer versions:
* 0.50.0: {'install_arg' in 'configure_file'}
NOTICE: Future-deprecated features used:
* 0.56.0: {'dependency.get_pkgconfig_variable', 'meson.build_root', 'meson.source_root'}
* 1.1.0: {'boolean option' keyword argument 'value' of type 'str'}

xserver 21.1.18

User defined options
  glamor      : true
  localstatedir : /var
  prefix      : /usr
  xkb_output_dir: /var/lib/xkb

Found ninja-1.13.1 at /usr/bin/ninja
[121/649] Compiling C object os/liblibxserver_os.a.p/rpcauth.c.o
In file included from ../os/rpcauth.c:47:
/usr/include/tirpc/rpc/rpc.h:83:12: warning: redundant redeclaration of 'bindresvport' [-Wredundant-decl
]
   83 | extern int bindresvport(int, struct sockaddr_in *);
       |             ^~~~~~
In file included from /usr/include/tirpc/rpc/rpc.h:40:
/usr/include/netinet/in.h:529:12: note: previous declaration of 'bindresvport' with type 'int(int, struct
sockaddr_in *)'
   529 | extern int bindresvport (int __sockfd, struct sockaddr_in *__sock_in) __THROW;
       |             ^~~~~~
[649/649] Linking target test/tests
-bash-5.3# ldconfig
-bash-5.3# ninja test
[0/1] Running all tests
1/6 xserver / modesetting symbol test      OK          0.05s
2/6 xserver:xvfb / XTS                     SKIP         0.06s   exit status 77
3/6 xserver / request-length               OK          0.34s
4/6 xserver / damage-primitives            OK          0.34s
5/6 xserver / sync                         OK          0.33s
6/6 xserver / unit                         OK          0.37s

Ok:          5
Fail:        0
Skipped:     1

Full log written to /root/sources/xorg-server-21.1.18/build/meson-logs/testlog.txt

```

Notes: no notes.

- Instalación de **libevdev-1.13.4** finalizada a las
  - El paquete libevdev contiene funciones comunes para los controladores de entrada de Xorg.
  - Necesario para Xorg Evdev Driver-2.11.0, libinput-1.29.0 y Xorg Synaptics Driver-1.10.0.

```

Source dir: /root/sources/libevdev-1.13.4
Build dir: /root/sources/libevdev-1.13.4/build
Build type: native build
WARNING: failed to process netrc file: Macro definition missing null line terminator. (/root/.netrc, line
5).
Project name: libevdev
Project version: 1.13.4
C compiler for the host machine: cc (gcc 15.2.0 "cc (GCC) 15.2.0")
C linker for the host machine: cc ld.bfd 2.45
Host machine cpu family: x86_64
Host machine cpu: x86_64
Library m found: YES
Library rt found: YES
Program libevdev/make-event-names.py found: YES (/root/sources/libevdev-1.13.4/libevdev/make-event-names.
py)
Configuring event-names.h with command
Found pkg-config: YES (/usr/bin/pkg-config) 2.5.1
Configuring libevdev.3 using configuration
Dependency check skipped: feature tests disabled
Program doxygen skipped: feature documentation disabled
Configuring config.h using configuration
Build targets in project: 6

libevdev 1.13.4

User defined options
  buildtype      : release
  documentation  : disabled
  prefix         : /usr
  tests          : disabled

Found ninja-1.13.1 at /usr/bin/ninja
[15/15] Linking target libevdev-tweak-device
-bash-5.3# ninja install
[0/1] Installing files
Installing libevdev.so.2.3.0 to /usr/lib64
Installing touchpad-edge-detector to /usr/bin
Installing mouse-dpi-tool to /usr/bin
Installing libevdev-tweak-device to /usr/bin
Installing /root/sources/libevdev-1.13.4/libevdev/libevdev.h to /usr/include/libevdev-1.0/libevdev
Installing /root/sources/libevdev-1.13.4/libevdev/libevdev-uinput.h to /usr/include/libevdev-1.0/libevdev
Installing /root/sources/libevdev-1.13.4/build/libevdev.3 to /usr/share/man/man3
Installing /root/sources/libevdev-1.13.4/tools/libevdev-tweak-device.1 to /usr/share/man/man1
Installing /root/sources/libevdev-1.13.4/tools/touchpad-edge-detector.1 to /usr/share/man/man1
Installing /root/sources/libevdev-1.13.4/tools/mouse-dpi-tool.1 to /usr/share/man/man1
Installing /root/sources/libevdev-1.13.4/build/meson-private/libevdev.pc to /usr/lib64/pkgconfig
Installing symlink pointing to libevdev.so.2.3.0 to /usr/lib64/libevdev.so.2
Installing symlink pointing to libevdev.so.2 to /usr/lib64/libevdev.so
-bash-5.3#

```

○

Notes: no notes.

- Instalación de **mtdev-1.1.7** finalizada a las
  - El paquete mtdev contiene la Biblioteca de Traducción del Protocolo Multitáctil, que se utiliza para transformar todas las variantes de eventos MT (Multitouch) del kernel al protocolo tipo B con ranuras.
  - Necesario para Xorg Evdev Driver-2.11.0 y libinput-1.29.0.

```

-----
Libraries have been installed in:
    /usr/lib

If you ever happen to want to link against installed libraries
in a given directory, LIBDIR, you must either use libtool, and
specify the full pathname of the library, or use the '-LLIBDIR'
flag during linking and do at least one of the following:
    - add LIBDIR to the 'LD_LIBRARY_PATH' environment variable
      during execution
    - add LIBDIR to the 'LD_RUN_PATH' environment variable
      during linking
    - use the '-Wl,-rpath -Wl,LIBDIR' linker flag
    - have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.
-----

/usr/bin/mkdir -p '/usr/include'
/usr/bin/install -c -m 644 ../include/mtdev-mapping.h ../include/mtdev-plumbing.h ../include/mtdev.h '/usr/include'
make[2]: Leaving directory '/root/sources/mtdev-1.1.7/src'
make[1]: Leaving directory '/root/sources/mtdev-1.1.7/src'
Making install in test
make[1]: Entering directory '/root/sources/mtdev-1.1.7/test'
make[2]: Entering directory '/root/sources/mtdev-1.1.7/test'
  /usr/bin/mkdir -p '/usr/bin'
  /bin/sh ../libtool --mode=install /usr/bin/install -c mtdev-test '/usr/bin'
libtool: install: /usr/bin/install -c .libs/mtdev-test /usr/bin/mtdev-test
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/root/sources/mtdev-1.1.7/test'
make[1]: Leaving directory '/root/sources/mtdev-1.1.7/test'
make[1]: Entering directory '/root/sources/mtdev-1.1.7'
make[2]: Entering directory '/root/sources/mtdev-1.1.7'
make[2]: Nothing to be done for 'install-exec-am'.
  /usr/bin/mkdir -p '/usr/lib/pkgconfig'
  /usr/bin/install -c -m 644 mtdev.pc '/usr/lib/pkgconfig'
make[2]: Leaving directory '/root/sources/mtdev-1.1.7'
make[1]: Leaving directory '/root/sources/mtdev-1.1.7'
-bash-5.3#

```

Notes: no notes.

- Instalación de **Xorg Evdev Driver-2.11.0** finalizada a las
  - El paquete Xorg Evdev Driver contiene un controlador de entrada genérico para Linux destinado al servidor Xorg. Maneja teclados, ratones, touchpads y dispositivos Wacom, aunque para un manejo avanzado de touchpads y Wacom se requieren controladores adicionales.

```
-----  
Libraries have been installed in:  
  /usr/lib/xorg/modules/input
```

If you ever happen to want to link against installed libraries in a given directory, LIBDIR, you must either use libtool, and specify the full pathname of the library, or use the '-LLIBDIR' flag during linking and do at least one of the following:

- add LIBDIR to the 'LD\_LIBRARY\_PATH' environment variable during execution
- add LIBDIR to the 'LD\_RUN\_PATH' environment variable during linking
- use the '-Wl,-rpath -Wl,LIBDIR' linker flag
- have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for more information, such as the ld(1) and ld.so(8) manual pages.

```
-----  
make[2]: Leaving directory '/root/sources/xf86-input-evdev-2.11.0/src'  
make[1]: Leaving directory '/root/sources/xf86-input-evdev-2.11.0/src'  
Making install in man  
make[1]: Entering directory '/root/sources/xf86-input-evdev-2.11.0/man'  
make[2]: Entering directory '/root/sources/xf86-input-evdev-2.11.0/man'  
make[2]: Nothing to be done for 'install-exec-am'.  
  /usr/bin/mkdir -p '/usr/share/man/man4'  
  /usr/bin/install -c -m 644 evdev.4 '/usr/share/man/man4'  
make[2]: Leaving directory '/root/sources/xf86-input-evdev-2.11.0/man'  
make[1]: Leaving directory '/root/sources/xf86-input-evdev-2.11.0/man'  
Making install in include  
make[1]: Entering directory '/root/sources/xf86-input-evdev-2.11.0/include'  
make[2]: Entering directory '/root/sources/xf86-input-evdev-2.11.0/include'  
make[2]: Nothing to be done for 'install-exec-am'.  
  /usr/bin/mkdir -p '/usr/include/xorg'  
  /usr/bin/install -c -m 644 evdev-properties.h '/usr/include/xorg'  
make[2]: Leaving directory '/root/sources/xf86-input-evdev-2.11.0/include'  
make[1]: Leaving directory '/root/sources/xf86-input-evdev-2.11.0/include'  
make[1]: Entering directory '/root/sources/xf86-input-evdev-2.11.0'  
make[2]: Entering directory '/root/sources/xf86-input-evdev-2.11.0'  
make[2]: Nothing to be done for 'install-exec-am'.  
  /usr/bin/mkdir -p '/usr/share/X11/xorg.conf.d'  
  /usr/bin/install -c -m 644 10-evdev.conf '/usr/share/X11/xorg.conf.d'  
  /usr/bin/mkdir -p '/usr/lib/pkgconfig'  
  /usr/bin/install -c -m 644 xorg-evdev.pc '/usr/lib/pkgconfig'  
make[2]: Leaving directory '/root/sources/xf86-input-evdev-2.11.0'  
make[1]: Leaving directory '/root/sources/xf86-input-evdev-2.11.0'  
-bash-5.3#
```

○

Notes: no notes.

- Instalación de **libinput-1.29.0** finalizada a las
  - libinput es una biblioteca que gestiona dispositivos de entrada para servidores de pantalla y otras aplicaciones que necesitan tratar directamente con dispositivos de entrada.

```

Ok:          1
Fail:        27

Full log written to /root/sources/libinput-1.29.0/build/meson-logs/testlog.txt
-bash-5.3# ninja install
[1/2] Installing files
Installing subdir /root/sources/libinput-1.29.0/quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/30-vendor-oracle.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/30-vendor-logitech.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/50-system-chicony.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/30-vendor-ite.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/50-system-starlabs.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/50-system-gigabyte.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/30-vendor-kensington.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/30-vendor-microsoft.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/50-system-asus.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/50-system-dell.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/50-system-graviton.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/50-system-huawei.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/30-vendor-vmware.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/30-vendor-cypress.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/50-system-lenovo.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/50-system-vaio.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/50-system-chuwi.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/30-vendor-a4tech.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/50-system-lg.quirks to /usr/share/libinput
Installing /root/sources/libinput-1.29.0/quirks/50-system-system76.quirks to /usr/share/libinput

```

Notes: estos errores se deben a que no se instalaron ciertas dependencias opcionales para los tests.

- Instalación de **xorg-libinput-1.5.0** finalizada a las
  - El controlador X.Org Libinput es un envoltorio ligero alrededor de libinput y permite que libinput se utilice para los dispositivos de entrada en X. Este controlador puede usarse como reemplazo directo de evdev y synaptics.

```

PASS: test-draglock
PASS: test-bezier
=====
Testsuite summary for xf86-input-libinput 1.5.0
=====
# TOTAL: 2
# PASS: 2
# SKIP: 0
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====
make[3]: Leaving directory '/root/sources/xf86-input-libinput-1.5.0/test'
make[2]: Leaving directory '/root/sources/xf86-input-libinput-1.5.0/test'
make[1]: Leaving directory '/root/sources/xf86-input-libinput-1.5.0/test'
make[1]: Entering directory '/root/sources/xf86-input-libinput-1.5.0'
make[1]: Leaving directory '/root/sources/xf86-input-libinput-1.5.0'
-bash-5.3#

```

Notes: make test succesful.

- Instalación de **Xorg Synaptics Driver-1.10.0** finalizada a las
  - El paquete Xorg Synaptics Driver contiene el controlador de entrada de X.Org, los programas de soporte y el SDK para touchpads Synaptics. Aunque el controlador evdev puede manejar touchpads bastante bien, este controlador es necesario si se



desean utilizar funciones avanzadas como el multitap, el desplazamiento mediante el touchpad, la desactivación del touchpad mientras se escribe, entre otras.

```
-----
make[2]: Leaving directory '/root/sources/xf86-input-synaptics-1.10.0/src'
make[1]: Leaving directory '/root/sources/xf86-input-synaptics-1.10.0/src'
Making install in man
make[1]: Entering directory '/root/sources/xf86-input-synaptics-1.10.0/man'
make[2]: Entering directory '/root/sources/xf86-input-synaptics-1.10.0/man'
make[2]: Nothing to be done for 'install-exec-am'.
/usr/bin/mkdir -p '/usr/share/man/man4'
/usr/bin/install -c -m 644 synaptics.4 '/usr/share/man/man4'
/usr/bin/mkdir -p '/usr/share/man/man1'
/usr/bin/install -c -m 644 synclient.1 '/usr/share/man/man1'
/usr/bin/mkdir -p '/usr/share/man/man1'
/usr/bin/install -c -m 644 syndaemon.1 '/usr/share/man/man1'
make[2]: Leaving directory '/root/sources/xf86-input-synaptics-1.10.0/man'
make[1]: Leaving directory '/root/sources/xf86-input-synaptics-1.10.0/man'
Making install in tools
make[1]: Entering directory '/root/sources/xf86-input-synaptics-1.10.0/tools'
make[2]: Entering directory '/root/sources/xf86-input-synaptics-1.10.0/tools'
/usr/bin/mkdir -p '/usr/bin'
/bin/sh ../libtool --mode=install /usr/bin/install -c synclient syndaemon '/usr/bin'
libtool: install: /usr/bin/install -c synclient /usr/bin/synclient
libtool: install: /usr/bin/install -c syndaemon /usr/bin/syndaemon
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/root/sources/xf86-input-synaptics-1.10.0/tools'
make[1]: Leaving directory '/root/sources/xf86-input-synaptics-1.10.0/tools'
Making install in conf
make[1]: Entering directory '/root/sources/xf86-input-synaptics-1.10.0/conf'
make[2]: Entering directory '/root/sources/xf86-input-synaptics-1.10.0/conf'
make[2]: Nothing to be done for 'install-exec-am'.
/usr/bin/mkdir -p '/usr/share/X11/xorg.conf.d'
/usr/bin/install -c -m 644 70-synaptics.conf '/usr/share/X11/xorg.conf.d'
make[2]: Leaving directory '/root/sources/xf86-input-synaptics-1.10.0/conf'
make[1]: Leaving directory '/root/sources/xf86-input-synaptics-1.10.0/conf'
make[1]: Entering directory '/root/sources/xf86-input-synaptics-1.10.0'
make[2]: Entering directory '/root/sources/xf86-input-synaptics-1.10.0'
make[2]: Nothing to be done for 'install-exec-am'.
/usr/bin/mkdir -p '/usr/lib/pkgconfig'
/usr/bin/install -c -m 644 xorg-synaptics.pc '/usr/lib/pkgconfig'
make[2]: Leaving directory '/root/sources/xf86-input-synaptics-1.10.0'
make[1]: Leaving directory '/root/sources/xf86-input-synaptics-1.10.0'
-bash-5.3#
```

Notes: no notes.

- Instalación de **twm-1.0.13.1** finalizada a las 14:36
  - El paquete twm-1.0.13.1 contiene un gestor de ventanas minimalista.
  - 
  - Crucial para el sistema X Window.

```

-bash-5.3# make install
Making install in src
make[1]: Entering directory '/root/sources/twm-1.0.13.1/src'
make install-am
make[2]: Entering directory '/root/sources/twm-1.0.13.1/src'
make[3]: Entering directory '/root/sources/twm-1.0.13.1/src'
  /usr/bin/mkdir -p '/usr/bin'
  /usr/bin/install -c twm '/usr/bin'
  /usr/bin/mkdir -p '/etc/X11/app-defaults'
  /usr/bin/install -c -m 644 system.twmrc '/etc/X11/app-defaults'
make[3]: Leaving directory '/root/sources/twm-1.0.13.1/src'
make[2]: Leaving directory '/root/sources/twm-1.0.13.1/src'
make[1]: Leaving directory '/root/sources/twm-1.0.13.1/src'
Making install in man
make[1]: Entering directory '/root/sources/twm-1.0.13.1/man'
make[2]: Entering directory '/root/sources/twm-1.0.13.1/man'
make[2]: Nothing to be done for 'install-exec-am'.
  /usr/bin/mkdir -p '/usr/share/man/man1'
  /usr/bin/install -c -m 644 twm.1 '/usr/share/man/man1'
make[2]: Leaving directory '/root/sources/twm-1.0.13.1/man'
make[1]: Leaving directory '/root/sources/twm-1.0.13.1/man'
make[1]: Entering directory '/root/sources/twm-1.0.13.1'
make[2]: Entering directory '/root/sources/twm-1.0.13.1'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/root/sources/twm-1.0.13.1'
make[1]: Leaving directory '/root/sources/twm-1.0.13.1'
-bash-5.3#

```

○

Notes: no notes.

- Instalación de **luit-20240910** finalizada a las
  - El paquete luit es un filtro que puede ejecutarse entre una aplicación cualquiera y un emulador de terminal UTF-8. Convierte la salida de la aplicación desde la codificación del locale a UTF-8, y la entrada del terminal desde UTF-8 a la codificación del locale.
  - Necesario para xterm-401.

```

-bash-5.3# make install
/usr/bin/install -c luit /usr/bin/`echo luit| sed 's,x,x,`
/bin/sh ./mininstall.sh "/usr/bin/install -c -m 644" ./luit.man /usr/share/man/man1/`echo luit| sed '
s,x,x,`.1 /usr /usr/share/X11/locale/locale.alias
/usr/bin/install -c -m 644 ./luit.man /usr/share/man/man1/luit.1
Completed installation of executables and documentation.
-bash-5.3#

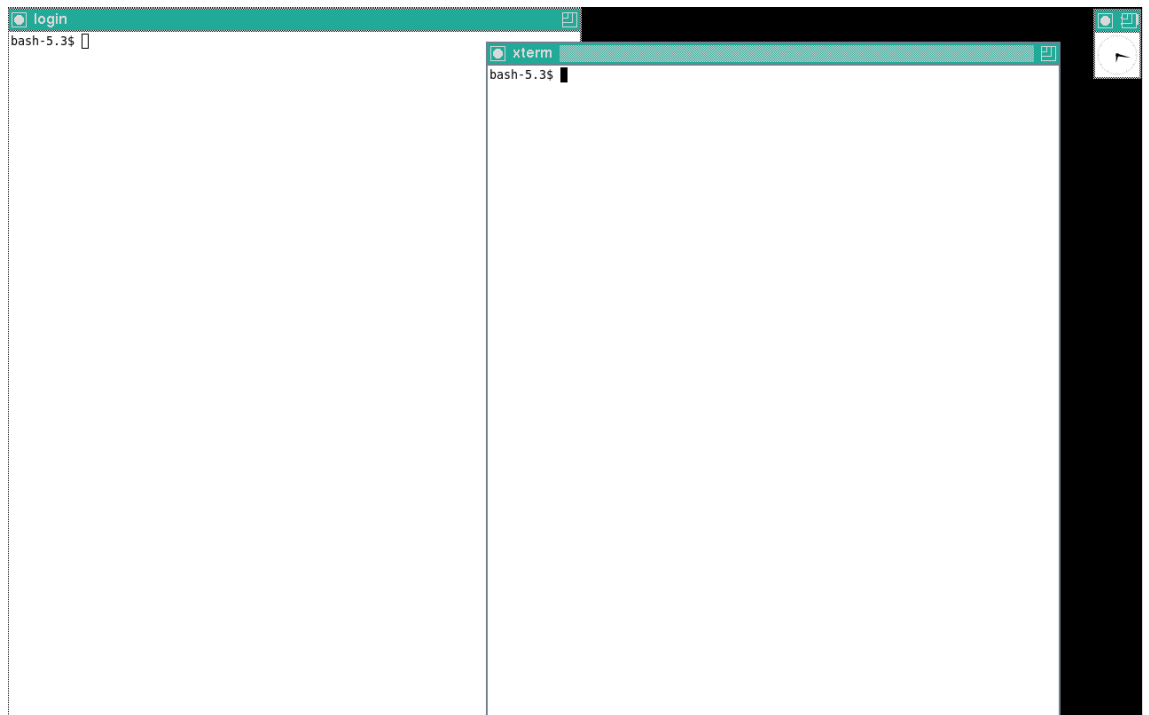
```

○

Notes: no notes.

- Instalación de **xterm-401** finalizada a las
  - xterm es un simulador de terminal para el sistema X Window.
  - Crucial para el sistema X Window.
  - No se tiene captura del proceso, por alguna razón.





Notes: fue necesario la instalación de un monospace TTF o OTF como Dejavu fonts.

- Instalación de **xclock-1.1.1** finalizada a las 14:41

- El paquete xclock contiene una aplicación de reloj simple que se utiliza en la configuración predeterminada de xinit.
- Completamente opcional.

```
-bash-5.3# make install
Making install in man
make[1]: Entering directory '/root/sources/xclock-1.1.1/man'
make[2]: Entering directory '/root/sources/xclock-1.1.1/man'
make[2]: Nothing to be done for 'install-exec-am'.
/usr/bin/mkdir -p '/usr/share/man/man1'
/usr/bin/install -c -m 644 xclock.1 '/usr/share/man/man1'
make[2]: Leaving directory '/root/sources/xclock-1.1.1/man'
make[1]: Leaving directory '/root/sources/xclock-1.1.1/man'
make[1]: Entering directory '/root/sources/xclock-1.1.1'
make[2]: Entering directory '/root/sources/xclock-1.1.1'
/usr/bin/mkdir -p '/usr/bin'
/usr/bin/install -c xclock '/usr/bin'
/usr/bin/mkdir -p '/etc/X11/app-defaults'
/usr/bin/install -c -m 644 app-defaults/XClock-color app-defaults/XClock '/etc/X11/app-defaults'
make[2]: Leaving directory '/root/sources/xclock-1.1.1'
make[1]: Leaving directory '/root/sources/xclock-1.1.1'
-bash-5.3#
```

Notes: no notes.

- Instalación de **xinit-1.4.4** finalizada a las

- El paquete xinit contiene un script utilizable para iniciar el servidor X.
- Built with Xorg Libraries, twm-1.0.13.1, xclock-1.1.1 y xterm-401.
- Crucial para el sistema X Window.
- Contiene startx, que inicializa una sesión X y contiene xinit, que es el inicializador del sistema X Window.

```

-bash-5.3# make install &&
> ldconfig
Making install in man
make[1]: Entering directory '/root/sources/xinit-1.4.4/man'
make[2]: Entering directory '/root/sources/xinit-1.4.4/man'
make[2]: Nothing to be done for 'install-exec-am'.
/usr/bin/mkdir -p '/usr/share/man/man1'
/usr/bin/install -c -m 644 startx.1 xinit.1 '/usr/share/man/man1'
make[2]: Leaving directory '/root/sources/xinit-1.4.4/man'
make[1]: Leaving directory '/root/sources/xinit-1.4.4/man'
make[1]: Entering directory '/root/sources/xinit-1.4.4'
make[2]: Entering directory '/root/sources/xinit-1.4.4'
/usr/bin/mkdir -p '/usr/bin'
/usr/bin/install -c xinit '/usr/bin'
/usr/bin/mkdir -p '/usr/bin'
/usr/bin/install -c startx '/usr/bin'
/usr/bin/mkdir -p '/etc/X11/app-defaults'
/usr/bin/install -c -m 644 xinitrc '/etc/X11/app-defaults'
make[2]: Leaving directory '/root/sources/xinit-1.4.4'
make[1]: Leaving directory '/root/sources/xinit-1.4.4'
-bash-5.3#

```

○

Notes: no notes.

- Instalación de **Xorg Legacy** finalizada a las 14:55
  - El antecesor de Xorg (X11R1, en 1987) inicialmente solo proporcionaba fuentes de mapas de bits, junto con una herramienta (bdfpc) para ayudar en su instalación. Con la introducción de xorg-server-1.19.0 y libXfont2, muchas personas ya no las necesitan. Sin embargo, todavía existen algunos paquetes antiguos que pueden requerirlas o beneficiarse de estas fuentes obsoletas, por lo que los siguientes paquetes se muestran aquí.
  - Paquetes instalados:
    - bdfpc-1.1.1
    - font-adobe-100dpi-1.0.4
    - font-adobe-75dpi-1.0.4
    - font-jis-misc-1.0.4
    - font-daewoo-misc-1.0.4
    - font-isas-misc-1.0.4
    - font-misc-misc-1.1.3

```

FINISHED --2026-01-19 14:48:43--
Total wall clock time: 6.1s
Downloaded: 7 files, 4.8M in 2.5s (1.93 MB/s)
bdfpc-1.1.1.tar.xz: OK
font-adobe-100dpi-1.0.4.tar.xz: OK
font-adobe-75dpi-1.0.4.tar.xz: OK
font-jis-misc-1.0.4.tar.xz: OK
font-daewoo-misc-1.0.4.tar.xz: OK
font-isas-misc-1.0.4.tar.xz: OK
font-misc-misc-1.1.3.tar.xz: OK
-bash-5.3# pwd
/root/sources/legacy
-bash-5.3# cd ..

```

○

Notes: no notes.

- Instalación de **libtirpc-1.3.6** finalizada a las 23:40
  - El paquete libtirpc contiene bibliotecas que brindan soporte a programas que utilizan la API de Llamadas a Procedimientos Remotos (RPC). Reemplaza a las entradas de RPC que antes estaban en glibc, pero no a las de la biblioteca NIS.
  - Necesario para libnsl-2.0.1.
  - Opcional para para Linux-PAM-1.7.1.

Notes: no notes.

- Instalación de **libnsl-2.0.1** finalizada a las 23:52
  - El paquete libnsl contiene la interfaz pública del cliente para NIS (YP). Reemplaza a la biblioteca NIS que antes formaba parte de glibc.
  - Opcional para para Linux-PAM-1.7.1.

```
-----
Libraries have been installed in:
  /usr/lib

If you ever happen to want to link against installed libraries
in a given directory, LIBDIR, you must either use libtool, and
specify the full pathname of the library, or use the '-LLIBDIR'
flag during linking and do at least one of the following:
- add LIBDIR to the 'LD_LIBRARY_PATH' environment variable
  during execution
- add LIBDIR to the 'LD_RUN_PATH' environment variable
  during linking
- use the '-Wl,-rpath -Wl,LIBDIR' linker flag
- have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.
-----
/usr/bin/mkdir -p '/usr/include/rpcsvc'
/usr/bin/install -c -m 644 rpcsvc/ypclnt.h rpcsvc/ypupd.h rpcsvc/yp_prot.h rpcsvc/yp.x rpcsvc/yp.h rpcsv
c/yppasswd.x rpcsvc/yppasswd.h '/usr/include/rpcsvc'
make[2]: Leaving directory '/root/sources/libnsl-2.0.1/src'
make[1]: Leaving directory '/root/sources/libnsl-2.0.1/src'
make[1]: Entering directory '/root/sources/libnsl-2.0.1'
make[2]: Entering directory '/root/sources/libnsl-2.0.1'
make[2]: Nothing to be done for 'install-exec-am'.
/usr/bin/mkdir -p '/usr/lib/pkgconfig'
/usr/bin/install -c -m 644 libnsl.pc '/usr/lib/pkgconfig'
make[2]: Leaving directory '/root/sources/libnsl-2.0.1'
make[1]: Leaving directory '/root/sources/libnsl-2.0.1'
-bash-5.3#
```

Notes: no notes.

- Instalación de **rpcsvc-proto-1.4.4** finalizada a las 23:57
  - El paquete rpcsvc-proto contiene los archivos de protocolo y cabeceras de rpcsvc, que antes se incluían con glibc y que no están presentes en la biblioteca de reemplazo libtirpc-1.3.6, junto con el programa rpcgen.
  - Opcional para Linux-PAM-1.7.1.

```

make[1]: Leaving directory '/root/sources/rpcsvc-proto-1.4.4/po'
make[1]: Entering directory '/root/sources/rpcsvc-proto-1.4.4'
make[2]: Entering directory '/root/sources/rpcsvc-proto-1.4.4'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/root/sources/rpcsvc-proto-1.4.4'
make[1]: Leaving directory '/root/sources/rpcsvc-proto-1.4.4'
-bash-5.3#

```

○

Notes: no notes.

20/01/2026

Instalación de **cracklib-2.10.3** finalizada a las 00:05

- El paquete CrackLib contiene una biblioteca utilizada para imponer contraseñas fuertes comparando las contraseñas elegidas por los usuarios con palabras de listas de palabras seleccionadas.
- Necesario para libpwquality-1.4.5.

```

-----
Libraries have been installed in:
  /usr/lib/python3.13/site-packages

If you ever happen to want to link against installed libraries
in a given directory, LIBDIR, you must either use libtool, and
specify the full pathname of the library, or use the '-LLIBDIR'
flag during linking and do at least one of the following:
- add LIBDIR to the 'LD_LIBRARY_PATH' environment variable
  during execution
- add LIBDIR to the 'LD_RUN_PATH' environment variable
  during linking
- use the '-Wl,-rpath -Wl,LIBDIR' linker flag
- have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.
-----

/usr/bin/mkdir -p '/usr/lib/python3.13/site-packages'
/usr/bin/install -c -m 644 cracklib.py test_cracklib.py '/usr/lib/python3.13/site-packages'
Byte-compiling python modules...
cracklib.pytest_cracklib.py
Byte-compiling python modules (optimized versions) ...
cracklib.pytest_cracklib.py
make[2]: Leaving directory '/root/sources/cracklib-2.10.3/python'
make[1]: Leaving directory '/root/sources/cracklib-2.10.3/python'
Making install in dicts
make[1]: Entering directory '/root/sources/cracklib-2.10.3/dicts'
make[2]: Entering directory '/root/sources/cracklib-2.10.3/dicts'
make[2]: Nothing to be done for 'install-exec-am'.
  /usr/bin/mkdir -p '/usr/share/cracklib'
  /usr/bin/install -c -m 644 cracklib-small '/usr/share/cracklib'
make[2]: Leaving directory '/root/sources/cracklib-2.10.3/dicts'
make[1]: Leaving directory '/root/sources/cracklib-2.10.3/dicts'
make[1]: Entering directory '/root/sources/cracklib-2.10.3'
make[2]: Entering directory '/root/sources/cracklib-2.10.3'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/root/sources/cracklib-2.10.3'
make[1]: Leaving directory '/root/sources/cracklib-2.10.3'
-bash-5.3# xzcat ../cracklib-words-2.10.3.xz \

```

○

Notes: no notes.

- Instalación de **libpwquality-1.4.5** finalizada a las 00:14
  - El paquete libpwquality proporciona funciones comunes para verificar la calidad de las contraseñas y también para puntuarlas según su aparente aleatoriedad. La biblioteca también ofrece una función para generar contraseñas aleatorias con buena pronunciabilidad.
  - Necesario para Linux-PAM-1.7.1 para forzar contraseñas seguras.

```
Looking in links: dist
Processing ./dist/pwquality-1.4.5-cp313-cp313-linux_x86_64.whl
Installing collected packages: pwquality
Successfully installed pwquality-1.4.5
-bash-5.3#
```

Notes: no notes.

- Instalación de **Linux-PAM-1.7.1** finalizada a las 01:01
  - El paquete Linux PAM contiene Módulos de Autenticación Conectables que el administrador del sistema local utiliza para controlar cómo los programas de aplicación autentican a los usuarios.
  - Built with libnsl-2.0.1, libtirpc-1.3.6, rpcsvs-proto-1.4.4, docbook-xml-5.0 y libpwquality-1.4.5.
  - La razón por la instalación de este paquete es que xinit necesita de Linux-PAM-1.7.1 debido a su registro de sesiones de usuarios.
  - Necesario para Shadow-4.18.0 y Systemd-257.8.

```
Ok:              78
Fail:            0

Full log written to /root/sources/Linux-PAM-1.7.1/build/meson-logs/testlog.txt
-bash-5.3#
```

```

-bash-5.3# install -vdm755 /etc/pam.d &&
> cat > /etc/pam.d/system-account << "EOF" &&
> # Begin /etc/pam.d/system-account
>
> account    required    pam_unix.so
etc/pam.>
> # End /etc/pam.d/system-account
> EOF
>
> cat > /etc/pam.d/system-auth << "EOF" &&
> # Begin /etc/pam.d/system-auth
>
> auth        required    pam_unix.so
>
> # End /etc/pam.d/system-auth
> EOF
>
> cat > /etc/pam.d/system-session << "EOF" &&
> # Begin /etc/pam.d/system-session
>
> session     required    pam_unix.so
>
> # End /etc/pam.d/system-session
> EOF
>
> cat > /etc/pam.d/system-password << "EOF"
> # Begin /etc/pam.d/system-password
>
> # use yescrypt hash for encryption, use shadow, and try to use any
> # previously defined authentication token (chosen password) set by any
> # prior module.
> password    required    pam_unix.so          yescrypt shadow try_first_pass
>
> # End /etc/pam.d/system-password
> EOF
-bash-5.3# ls /etc/pam.d
system-account  system-auth  system-password  system-session
-bash-5.3#

```

○

```

-bash-5.3# mv /etc/pam.d/system-password{,.orig} &&
> cat > /etc/pam.d/system-password << "EOF"
> # Begin /etc/pam.d/system-password
>
> # check new passwords for strength (man pam_pwquality)
> password required pam_pwquality.so authtok_type=UNIX retry=1 difok=1 \
> minlen=8 dcredit=0 ucredit=0 \
> lcredit=0 ocredit=0 minclass=1 \
> maxrepeat=0 maxsequence=0 \
> maxclassrepeat=0 gecheck=0 \
> dictcheck=1 usercheck=1 \
> enforcing=1 badwords="" \
> dictpath=/usr/lib/cracklib/pw_dict
>
> # use yescrypt hash for encryption, use shadow, and try to use any
> # previously defined authentication token (chosen password) set by any
> # prior module.
> password required pam_unix.so yescrypt shadow try_first_pass
>
> # End /etc/pam.d/system-password
> EOF
-bash-5.3# cd ..
-bash-5.3# pwd
/root/sources
-bash-5.3# cd Linux-PAM-1.7.1
-bash-5.3# cd build
-bash-5.3# cat > /etc/pam.d/other << "EOF"
> # Begin /etc/pam.d/other
>
> auth required pam_warn.so
> auth required pam_deny.so
> account required pam_warn.so
> account required pam_deny.so
> password required pam_warn.so
> password required pam_deny.so
> session required pam_warn.so
> .so
# > session required pam_deny.so
>
> # End /etc/pam.d/other
> EOF
-bash-5.3#

```

Notes: Shadow 4.18.0 y Systemd 257.8 deben reinstalarse y reconfigurarse después de instalar y configurar Linux PAM. Se tuvo que volver a configurar el kernel para la instalación de este paquete.

- Instalación de **Shadow-4.18.0** finalizada a las 09:29
  - Con Linux-PAM instalado, reinstalar Shadow permitirá que programas como login y su utilicen PAM.
  - Necesario para Systemd-257.8.

shadow will be compiled with the following features:

|                                |     |
|--------------------------------|-----|
| auditing support:              | no  |
| PAM support:                   | yes |
| suid account management tools: | no  |
| SELinux support:               | no  |
| Btrfs support:                 | yes |
| ACL support:                   | yes |
| Extended Attributes support:   | yes |
| tcb support (incomplete):      | no  |
| shadow group support:          | yes |
| S/Key support:                 | no  |
| SHA passwords encryption:      | yes |
| bcrypt passwords encryption:   | yes |
| yescrypt passwords encryption: | yes |
| nscd support:                  | yes |
| sssd support:                  | yes |
| subordinate IDs support:       | yes |
| enable lastlog:                | no  |
| enable logind:                 | yes |
| use file caps:                 | no  |
| install su:                    | yes |
| enabled vendor dir:            |     |

○

Notes: no notes.

- Instalación de **systemd-257.8** finalizada a las 13:01
  - El paquete systemd debe recompilarse con Linux-PAM instalado para proporcionar un servicio systemd-logind funcional, el cual ofrece muchas funciones adicionales para los paquetes dependientes.



```

-bash-5.3# grep 'pam_systemd' /etc/pam.d/system-session ||
> cat >> /etc/pam.d/system-session << "EOF"
> # Begin Systemd addition
>
> session required    pam_loginuid.so
> session optional    pam_systemd.so
>
> # End Systemd addition
> EOF
-bash-5.3# cat > /etc/pam.d/systemd-user << "EOF"
> # Begin /etc/pam.d/systemd-user
>
> account required    pam_access.so
> account include      system-account
>
> session required    pam_env.so
> session required    pam_limits.so
> session required    pam_loginuid.so
> session optional    pam_keyinit.so force revoke
> session optional    pam_systemd.so
>
> auth required       pam_deny.so
> password required    pam_deny.so
>
> # End /etc/pam.d/systemd-user
> EOF
-bash-5.3# systemctl daemon-reexec
-bash-5.3#

```

○

```

-bash-5.3# systemctl --version
systemd 257 (257.8)
+PAM -AUDIT -SELINUX -APPARMOR +IMA +IPE +SMACK -SECCOMP +GCRYPT -GNUTLS +OPENSSL +ACL +BLKID -CURL -ELF
UTILS -FIDO2 +IDN2 -IDN -IPTC +KMOD -LIBCRYPTSETUP -LIBCRYPTSETUP_PLUGINS +LIBFDISK +PCRE2 +PWQUALITY -P
11KIT -QRENCODE -TPM2 +BZIP2 +LZ4 +XZ +ZLIB +ZSTD -BPF_FRAMEWORK -BTF -XKBCOMMON +UTMP +SYSVINIT +LIBAR
HIVE
-bash-5.3# journalctl --version
systemd 257 (257.8)
+PAM -AUDIT -SELINUX -APPARMOR +IMA +IPE +SMACK -SECCOMP +GCRYPT -GNUTLS +OPENSSL +ACL +BLKID -CURL -ELF
UTILS -FIDO2 +IDN2 -IDN -IPTC +KMOD -LIBCRYPTSETUP -LIBCRYPTSETUP_PLUGINS +LIBFDISK +PCRE2 +PWQUALITY -P
11KIT -QRENCODE -TPM2 +BZIP2 +LZ4 +XZ +ZLIB +ZSTD -BPF_FRAMEWORK -BTF -XKBCOMMON +UTMP +SYSVINIT +LIBAR
HIVE

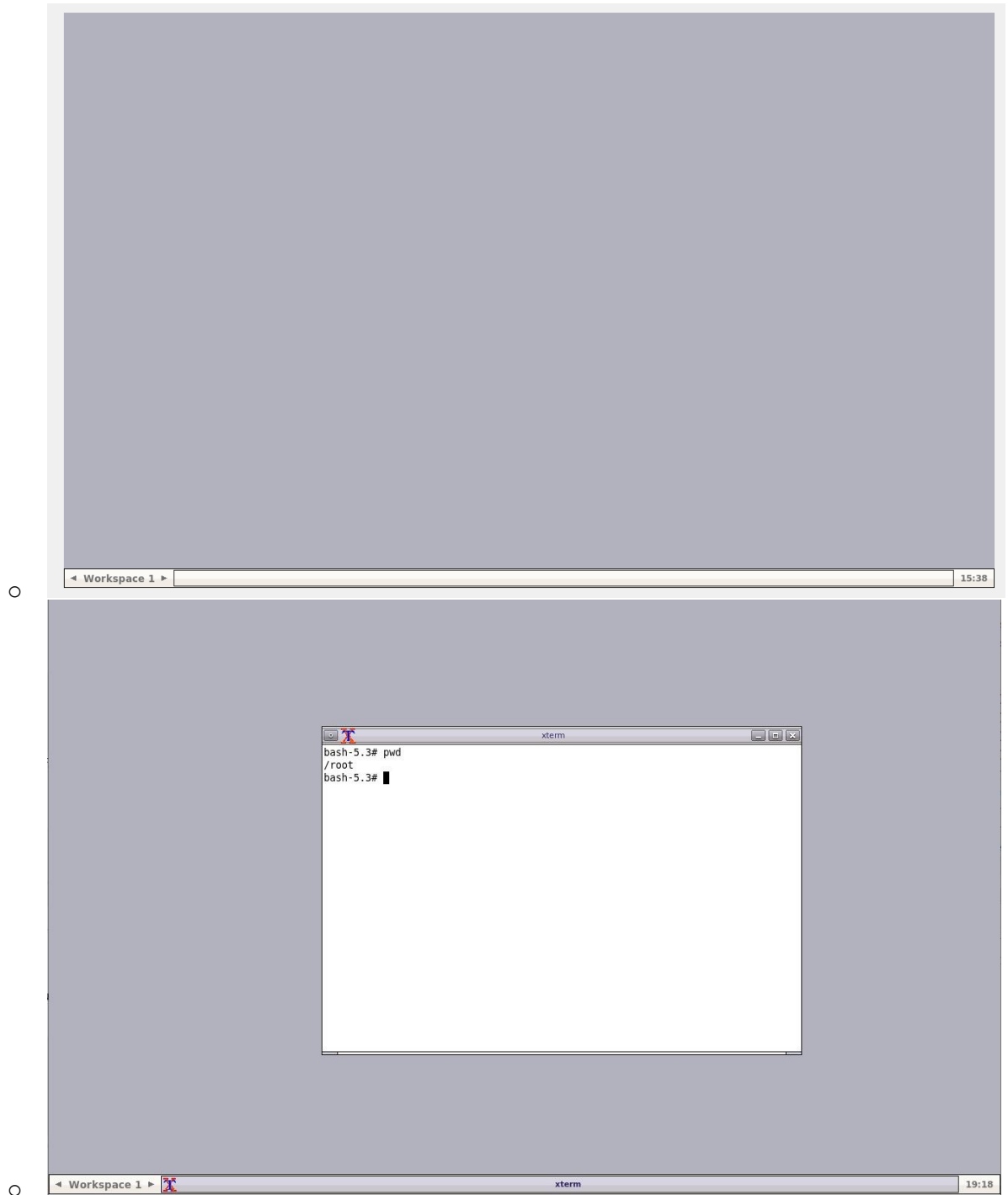
```

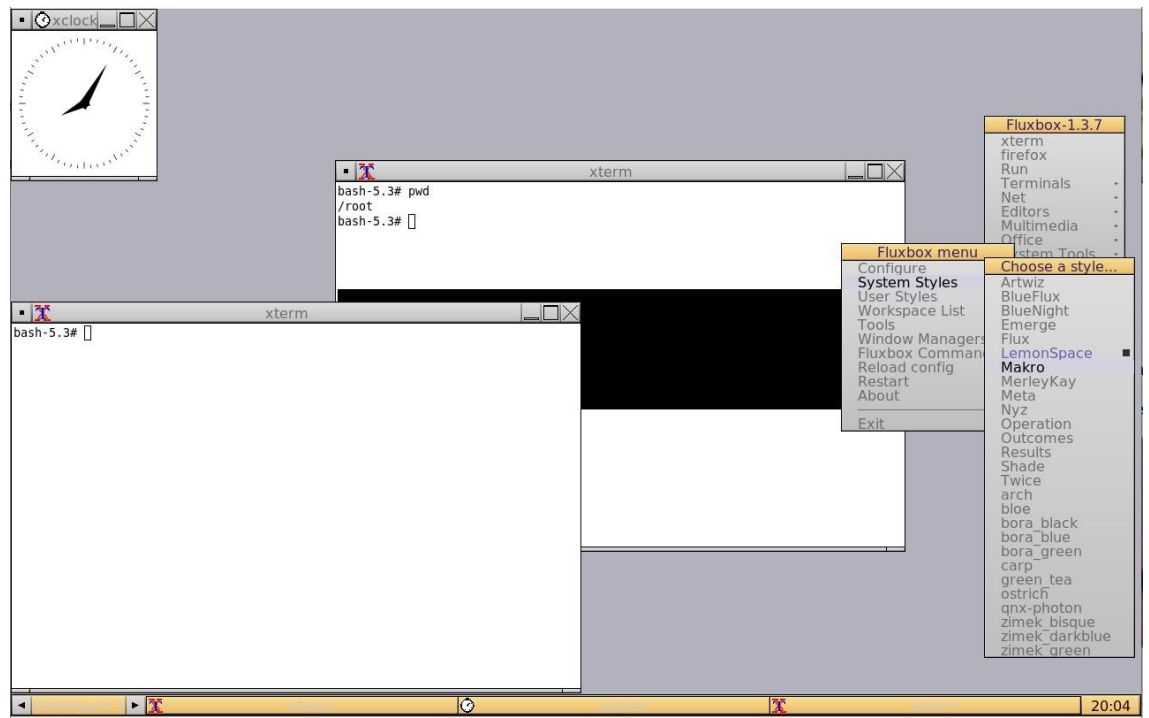
○

Notes: no notes.

## Capítulo 27 – Window Managers

- Instalación de **Fluxbox-1.3.7** finalizada a las 17:57
  - El paquete Fluxbox contiene un gestor de ventanas.





Notes:

- No aparece el cursor.